



**UNIVERSITY OF BOTSWANA
FACULTY OF SCIENCE
DEPARTMENT OF COMPUTER SCIENCE**

FINAL YEAR PROJECT PROPOSAL

KaboDitsha: A Web based Land Management System

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27 April 2026

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CHAPTER 1: INTRODUCTION

1.1 Background

Land is an important resource for economic development, cultural identity and social stability in Botswana. Land administration is handled by the twelve Main Land Boards and their subordinate Land Boards, operating under the Ministry of Lands and Agriculture and also governed by the Tribal Land Act(Cap 32:2). The Land Boards serve as the primary mechanism through which Batswana can get the residential plots in the cities, towns, villages and agricultural land.

The current land allocation policy, revised in 2019, states that a citizen is eligible for one plot per settlement type, specifically one plot in the city, one in a town or village and one farm plot. This policy ensures that land is distributed fairly among citizens and prevents cases where citizens are allocated multiple plots in a single settlement type. However, the implementation of this policy remains heavily dependent on manual and paper based processes that have persisted over the years.

Despite the Vision 2036 commitment to digital transformation and the government's broader e-governance initiatives to digitise services, the land application and allocation has not undergone significant modernisation. The current system heavily relies on a manual paper based approach, which has led to inefficiencies like prolonged processing times, administrative backlogs and limited transparency for applicants. This highlights the urgent need to modernise this process.

1.1.1 Current Digital Systems in Land Administration

Botswana's land administration currently relies on a fragmented digital landscape with multiple systems serving different purposes. For State Land (urban areas), the Department of Lands utilizes the State Land Integrated Management System (SLIMS) to manage waiting lists and allocations. However, parliamentary reports indicate that SLIMS has faced significant inefficiencies, prompting the government to transition to a new Land Information System (LIS) (Africa-Press, 2025a).

The transition has not been smooth. In Selebi Phikwe, allocations were halted in 2024 due to "failing of State Land Integrated Management Systems as a result of migration to the Land Information System" (Africa-Press, 2025a). Similarly, applicants who accepted transfers to Francistown in 2022 had allocations paused because "a decision was made that all land allocations should be made through the Land Information System (LIS) to address the

inefficiencies in the SLIMS" (Africa-Press, 2025a). Data migration challenges have delayed allocations, with the government aiming to resolve issues by December 2025.

For Tribal Land, which falls under the jurisdiction of 12 Main Land Boards and their subordinate boards as established by the Tribal Land Act (Cap 32:02) (Government of Botswana, 1968), a system called TLIMS (Tribal Land Information Management System) was developed years ago to assist with internal administration. However, its effectiveness has been limited by resource constraints and understaffing, "in most sub-land boards, there is only one person per cadre," leading to processing delays (Africa-Press, 2025b).

The Digital Services Bill, 2025, recently passed by Parliament, now mandates that public bodies, including Land Boards, "develop accessible digital platforms for the provision of digital services" (Daily News, 2025a). This legislative framework requires:

- Digital platforms accessible to citizens
- Online identity verification through a unique identifier
- Secure electronic payment mechanisms
- Compliance with national standards and interoperability protocols (Daily News, 2025b)

Despite these mandates, current systems remain internal, staff-facing tools not designed for citizen self-service. Applicants cannot apply online, upload documents, or track applications in real-time.

1.1.2 Emerging Citizen-Facing Initiatives

During the course of this research, an emerging government initiative was identified: the Land Applications KYC (LAKYC) portal (<https://lakyc.gov.bw>). This system, developed by the Geospatial Information Center under the Ministry of Lands and Agriculture, appears to be an attempt to digitize citizen identity verification for land applications (Geospatial Information Center, 2026).

However, preliminary observation reveals several limitations:

- **KYC-Only Focus:** The system appears primarily designed for identity verification rather than end-to-end application management. Users can update profiles and verify next of kin, but core functionality such as submitting new applications, tracking waiting list positions, or receiving status notifications is not evident from the interface.
- **Usability Challenges:** The interface exhibits design inconsistencies, including optional fields that functionally block dashboard access until completed and unclear messaging around application status. Fields marked "optional" behave as required, creating user friction.
- **Data Integration Gaps:** A personal test revealed that an existing land application was not automatically retrieved from Land Board databases, requiring the user to "report it missing", a manual workaround that suggests incomplete integration between the KYC system and underlying Land Board records.
- **Development Status:** The system was not publicly accessible during the initial research phase of this project, highlighting the nascent stage of Botswana's citizen-facing land digitization efforts.

The emergence of LAKYC confirms that the government recognizes the need for digital land services. However, the identified limitations reinforce the gap that KoboDitsha addresses: a comprehensive, user-centered platform for the entire land application lifecycle, from submission through allocation, with integrated waiting list management and transparent tracking.

1.2 Problem Statement

The land allocation process varies significantly in practice across regions and often involves a manual multi step procedure. Applicants are required to obtain application forms from the offices and pay the application fee of P50 in person. They are then required to submit the filled application form either through postal mail or hand delivery, resulting in immediate accessibility barriers for those without transport or postal access. Some land boards only accept hand delivery, others only postal mail while others do both, this results in immediate accessibility barriers.

Once received, the application forms are manually processed by the land board staff, who enter the applicant details into the digital waiting list after cross referencing the applicant details against the certified copy of Omang. The current system is designed to prevent multiple allocations through the one plot per settlement type policy but the verification process is manual and often has oversights. This takes a lot of time and is error prone. A recent report showed that approximately 1.4 million applicants were recorded across all land boards, the vetting process revealed that only 500 000 applications were genuinely eligible meaning that 64% of the records were either duplicates, already allocated applicants or those that did not meet the eligibility criteria (Daily News, 2022b). When Bobonong Sub-Land Board had 200 plots available for allocation, the vetting process had over 1000 interviewees, with many being turned down because they had already been allocated plots by other land boards for the same settlement type (Daily News, 2022a). Staff spent considerable amounts of time managing these flawed records instead of focusing on validating and making allocation decisions, while the high error rate undermines public trust in the service delivery of the Land boards.

Processing of applications varies by regions, with waiting times ranging from months to decades. The Bobonong Sub-Land Board, for instance, had a waiting list standing at 10 900 applicants in 2022, with the land board still processing applications dating back to 2005 which is a 17 year old backlog (Daily News, 2022a). In extreme cases, some applicants have passed away while waiting for allocation, highlighting a failure in timely service delivery. In Shashe West constituency, some residents applied for land in 1993/94 and only received

allocations in 2022 (Mmegi Online, 2025). The Minister of Lands and Agriculture acknowledged "disturbing reports which involved citizens who applied for land 30 years ago still waiting" (Mmegi Online, 2025).

Throughout the waiting period, applicants have no means of tracking their applications in real time. To inquire about their position in the queue, they must either visit the land board offices physically or attempt telephone contact, with no guarantee of the calls being answered or receiving accurate information. This creates additional administrative burdens on the staff as they spend time responding to these queries.

Recent government initiatives, such as the Land Applications KYC portal, indicate a recognition of the need for digitization. However, these efforts remain incomplete and fragmented. The LAKYC system focuses narrowly on identity verification, lacks full application lifecycle management and exhibits usability challenges including forced field completion and unclear status tracking. Crucially, existing land applications are not automatically integrated, requiring citizens to manually "report missing" submissions, a process that mirrors the very opacity the system aims to solve (Geospatial Information Center, 2026).

When decisions are finally made, either for allocation, rejection or duplicate detection, notifications are sent via postal mail, phone calls or sometimes both, often after significant delays. These forms of communication are unreliable as applicants may change their postal addresses, or their phone numbers, meaning that they may never receive these updates. Duplicate allocations require corrective interventions through committee's manual reconciliation which then extend the waiting period and service delivery.

1.3 Project Motivation

This project is motivated by a combination of personal experience, observed systematic inefficiencies and alignment with the national developmental priorities.

Having undergone the process myself, I experienced first hand frustrations of a non transparent and manually driven system. After applying just before turning 18, I was informed after two months via a phone call that my application was unsuccessful due to age eligibility, a delay that exposed lack of real time validation. Despite being placed on the waiting list with thousands of other Batswana after re-applying, I remained without a means of checking my place in the queue other than calling or visiting the offices in person for

inquiries. This exposure to physical access barriers helped me realise the frustrations that Botswana are facing.

During the course of this project, an emerging government portal (LAKYC) was identified, confirming that digitization efforts are underway. However, firsthand experience with this system revealed persistent gaps: a missing application that should have been retrieved automatically and interface friction that blocks dashboard access until all fields, including supposedly optional ones, are completed. These observations reinforce the need for a thoughtfully designed, integrated solution.

On a systematic level, Botswana's land allocation process remains paper based, has postal dependencies and manual record keeping. As evidenced in the problem statement, these practices result in prolonged processing times, high administrative overheads and lack of transparency. The absence of a centralised digital platform excludes vulnerable populations from equitable access to public service, as those in remote areas or without means to travel repeatedly or make phone calls are left out by the system. Furthermore, the manual system is prone to errors in duplicate detection and status updates, leading to backlogs, misallocations and public distrust.

Beyond addressing these inefficiencies, the project aligns with Botswana's Vision 2036 which emphasises digital transformation, improved public service delivery and inclusive development. The government has made a commitment to e-governance across all sectors, most importantly the Digital Services bill of 2025, classifies land boards as public bodies and legally mandates them to provide accessible electronic platforms for service delivery, accept digital submissions, issue approvals electronically and maintain records in digital format. This project contributes to the broader national agenda for modernising public services and responds to an emerging legal imperative for Land Board digitization.

1.4 Proposed Solution

To address these inefficiencies, this project proposes KaboDitsha, a web based platform that brings Botswana's land application process into the modern era.

KaboDitsha will provide citizens with an online portal where they can submit land applications, upload relevant documents such as their certified Omang and view their status on the waiting list. This eliminates the need for mailing forms or repeated office visits. Land Board staff will have access to a centralised dashboard to manage applications, verify

documents, update waiting lists and send instant notifications via the email. To further improve the experience, the platform may also include an AI-powered virtual assistant as an optional enhancement, time permitting.

By digitizing the process and increasing transparency, KaboDitsha will reduce delays, improve communication and help align Botswana's public services with current digital standards.

1.5 Project Objectives

The primary objective is to develop a platform that transforms land allocation from a paper-based system into a fast, transparent, reliable and accessible digital service. Specific objectives are:

- To digitise the entire land application process: citizens will be able to apply online, upload documents and receive immediate confirmation of submission.
- To implement real time tracking of applications: applicants can log into a secure dashboard, view their position on the waiting list and receive automatic updates via email and in-app notifications, eliminating the need for office visits.
- To develop an administration management system: staff can process applications, verify documents, update records and communicate with applicants through a single interface.
- To ensure the system is accessible and secure: the platform will be user intuitive/friendly, function on any device and implement authentication using email and password hashing, as well as role based access to ensure applicants and administrators each see a dashboard tailored to their permissions and responsibilities.

An AI-powered virtual assistant was identified as a potential enhancement but was not implemented due to time constraints. This remains a documented direction for future work to demonstrate the potential of conversational AI in public service delivery.

1.6 Project Scope

1.6.1 In Scope

The following are included within the scope of the project:

- Public Homepage (accessible to all visitors, serving as the 'Before You Apply' information hub) that shows: waiting list statistics per Land Board and Sub-Land

Board; a clear explanation of the one-plot-per-settlement-type rule; eligibility criteria and application process overview.

- Public Web Portal for Citizens where users can: create accounts and log in securely; submit land applications online; upload scanned copies of required documents such as Omang; check their position on the waiting list; and receive notifications about application updates via email and in-app notifications.
- Admin dashboard for Land Board staff to: review, verify and process applications; manually update application status, waiting list positions and backlog displays; manually flag and review potential duplicate applications based on the settlement type; and generate basic reports about applications.
- GPS-powered location features(with user permission) to identify the user's Land Board jurisdiction and check proximity to services such as roads, schools and clinics, etc, to the Land Board they want to apply to.
- User authentication, session handling and role based access control ensuring applicants and administrators each access functionality tailored to their permissions.

1.6.2 Out Of Scope

Due to the time constraints, cost considerations and the academic nature of this project, the following are explicitly excluded:

- Payment gateway integration of application fees
- AI assistant as core requirement: The AI chatbot was not implemented in the beta version. It remains a documented future enhancement for post-beta development.
- Fully automating approvals for land allocation
- Integration with existing national databases
- Multi language support beyond english
- Automated duplicate detection across all land boards
- Advanced predictive analysis for land allocation timelines
- SMS notifications (email and in-app notifications will be the primary communication channels)
- Offline functionality for the platform

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing literature and systems relevant to the development of KaboDitsha. The review covers digital land management systems, e-governance initiatives in Africa, role based access control mechanisms, AI-powered virtual assistants for public services, open source technologies for government systems and case studies of similar transformation projects. By examining these areas, this chapter establishes the theoretical and practical foundations of the proposed system and identified the gaps that KaboDitsha aims to address.

2.2 Digital land management systems

2.2.1 Evolution of land administration systems

The land administration system includes the process of recording and disseminating information about ownership, value and the use of land. Traditionally these systems were manual paper-based and included record keeping, physical storage of records and in-person transactions. The United Nations Economic Commission for Europe (UNECE, 1996) defines land administration as including land tenure, land value, land use and land development, all of which require accurate, accessible and secure information systems. Williamson et al. (2010) further argues that effective land administration is important in economic development as it enables secure property ownership, facilitates land markets and supports infrastructure planning.

The digitization has been driven by the following factors: the need for improved transparency and reduced corruption; the demand for faster processing times; the need for better data integrity and disaster recovery; and the pressure to reduce administrative overheads. Countries that have successfully digitized land administration report significant improvements in service delivery, reduced transaction times and enhanced public trust. The Food and Agriculture Organization (FAO, 2018) emphasises that digital governance systems must be integrated, interoperable and accessible to all stakeholders to achieve these benefits.

However, the literature also highlights common implementation problems including resistance to change among staff already accustomed to manual processes, concerns about data security and privacy and the digital divide that may exclude certain populations from accessing online services. The World Bank (2019) notes that successful digital land administration requires not just technology but also legal reform, capacity building and sustained political commitment.

2.2.2 Land Information Management Systems (LIMS)

Land Information Management systems are specialised software platforms designed to manage spatial data (the 'where') and non spatial data (the 'what') related to land parcels, ownership, transactions and associated legal documents. Modern Land Information Management systems include components for document mapping, workflow automation, payment processing and public access portals. Williamson et al. (2010) argues that the effective systems must include both the legal and the spatial frameworks to provide complete land information services.

A review of LIMS implementations shows that the front-end user interfaces and back-end data management should be clearly separated to allow iterative improvements without disrupting operations. It also shows that the mobile responsive design is critical as many users in developing places access government services primarily through their smartphones. The review also states that offline functionality is important for contexts where internet connectivity is unreliable. Lastly, it states that systems should include mechanisms for data validation and error detection at the point of entry to maintain data quality (UNECA 2017).

However, many LIMS implementations in Africa have faced sustainability changes. Studies of land administration projects on East Africa discovered that systems often fail after the initial project funding ends due to inadequate budgets for maintenance, lack of in-house technical expertise and insufficient training for end users (Ochieng & Wechuli, 2021). These findings underscore the importance of designing systems that are not only technically sound but also institutionally and financially sustainable.

2.2.3 Waiting list management in public services

Waiting list management is a critical component of service delivery in resource constrained environments where demand exceeds supply. Research in operations management and public administrations shows several principles for effective waiting list systems;

transparency regarding queue positions and expected wait times, fairness in allocation process to avoid queue jumping or favoritism, efficiency in processing to minimise overall wait times and communication with applicants to manage expectations and maintain trust.

Digital waiting list systems offer advantages over manual systems. They can provide real time status updates accessible remotely, automate notifications when an applicant's turn approaches, generate analytics to identify bottlenecks and optimize processing and maintain audit trails to ensure accountability. Studies from healthcare and house allocation contexts show that transparent waiting list systems significantly improve public satisfaction even when wait times remain long.

However, the literature also cautions that digital waiting lists must be designed with integrity protections to prevent manipulation. This includes secure authentication to verify applicant's identities, audit logging of all status changes, regular data reconciliation to detect anomalies and clear policies regarding how prioritization or reordering occurs. These safeguards help prevent corruption or fraud.

2.3 E-Governance in African contexts

2.3.1 E-Governance maturity models

UNESCO (2005) identifies four e-government stages: emerging (basic online presence), enhanced (more information and downloadable forms), transactional (two-way interaction with secure transactions) and connected (integrated services across departments). Most African countries remain in emerging or enhanced stages with exceptions like Rwanda, Mauritius and Estonia (Estonian Government, 2020).

Botswana's Digital Transformation Strategy 2021-2025 (Government of Botswana, 2021) aims to move toward transactional and connected stages, though land administration has lagged. The Digital Services Bill of 2025 (Government of Botswana, 2025) mandates public bodies to provide accessible electronic platforms; critical success factors include funding, capacity building, legal frameworks and inclusive design.

2.3.2 Trust and Data Privacy in Digital Government

Trust requires confidence in data security, system reliability and fair processing (Bwalya & Mutula, 2016; Carter & Bélanger, 2005). Botswana's Data Protection Act (2018) provides legal frameworks, but implementation requires encryption, access controls, audit logging and

secure authentication. Citizens are more likely to trust digital services when privacy policies are clearly communicated and value is demonstrated.

2.4 Role Based Access Control (RBAC) Systems

2.4.1 RBAC Principles and Models

RBAC restricts system access based on organisational roles with permissions associated to roles rather than individual users (Sandhu et al., 1996). NIST defines four levels: Flat, Hierarchical, Constrained and Symmetric RBAC (Ferraiolo et al., 2001). For public sector applications, flat or hierarchical RBAC typically suffices. Implementation involves authentication, authorization and session management.

2.4.2 RBAC in Government Information Systems

Government systems require granular access controls due to data sensitivity and regulatory requirements. Best practices include defining roles based on actual job functions, implementing least privilege, separating duties and regular auditing (Botha & Eloff, 2001). For land administration, typical roles include public applicants, landboard staff (clerks and managers) and administrators. Role-appropriate interfaces improve usability for users with limited technical expertise (Shneiderman et al., 2016).

2.5 Open source technologies for Government systems

2.5.1 Benefits and challenges of open-source in public sector

Open-source software (OSS) offers cost savings through eliminated licensing fees, transparency via publicly auditable code, flexibility without vendor restrictions, avoidance of vendor lock-in and community support (Open Source Initiative, 2007). These benefits make OSS particularly suitable for public sector applications requiring transparency and long-term sustainability.

However, successful adoption requires adequate technical capacity, evaluation of project maturity and sustainability and verification of integration with existing systems (Wheeler, 2015). Despite challenges, many governments have implemented large scale open-source initiatives. Estonia's X-Road data exchange platform, built on open source technologies, underpins the country's digital government infrastructure and has been adopted by other nations (Estonian Government, 2020). The Open Government Partnership encourages

member countries, including Botswana to adopt open source solutions (Open Government Partnership, 2019).

2.5.2 Open-Source stack for web applications

Modern web development has converged on mature open source technologies. The MERN Stack (MongoDB, Express.js, React, Node.js) has become industry standard for scalable web applications. React provides component based architecture for complex user interfaces with distinct roles (React Team, 2024). PostgreSQL offers enterprise grade relational database features including ACID (Atomicity, Consistency, Isolation and Durability) compliance and complex queries (PostgreSQL Documentation, 2024).

For geospatial applications, Leaflet.js is the leading open source JavaScript library for interactive maps (Leaflet Documentation, 2024). PostGIS extends PostgreSQL with spatial data types and functions, enabling sophisticated geographic queries without separate GIS software (PostGIS Documentation, 2024). These technologies have been successfully deployed in land administration systems worldwide.

Comprehensive documentation, active community forums and extensive third party libraries reduce development time. The large talent pool familiar with these technologies facilitates finding qualified developers and maintaining systems over time.

2.6 AI Powered virtual assistants for public services

Conversational AI systems have become increasingly prevalent in public sector applications, evolving from early rule-based chatbots to modern large language models (LLMs) capable of contextually aware, multi-turn conversations (Adamopoulou & Moussiades, 2020; Brown et al., 2020). For government services, key success factors include response accuracy, natural conversation flow and appropriate escalation to human agents for complex issues (Androutsopoulou et al., 2019). Where responses must be grounded in official sources, retrieval-augmented generation (RAG) offers a practical approach by retrieving relevant policy documents before generating answers (Lewis et al., 2020). Open-source models and local deployment frameworks make this feasible without recurring API costs (Touvron et al., 2023).

2.7 Case Studies of Digital Land Administration in Africa

2.7.1 Rwanda Land tenure regularization

Rwanda's Land tenure regularization program (2009-2014) registered over 11 million land parcels, providing formal documentation to millions of rural households through a digital Land Administration Information System (LAIS) (Ngoga, 2019).

Key lessons include combining field data collection with centralised digital records, extensive community engagement for buy-in and accuracy, strong political will and adequate funding and system accessibility at district offices nationwide (Sagashya & English, 2009). Ongoing challenges include system maintenance, capacity constraints in some districts and keeping records updated (Dushimimana, 2020). These lessons inform KoboDitsha's design regarding decentralised access and institutional support.

2.7.2 Kenya's Ardhi Sasa Platform

Kenya's Ardhi Sasa ("Land Now") platform digitizes land transactions including searches, applications and payments, aiming to reduce transaction time from months to days and minimize corruption by digitizing and tracking all process stages (Ochieng & Wechuli, 2021).

Successes include improved transparency as users can track applications online and receive SMS updates and significantly reduced transaction times. Challenges include uneven implementation across regions, with some registries lacking internet connectivity or staff training and complex user interfaces for first time users (Kariuki, 2020). The Kenya case underscores the importance of user centered design and parallel investments in infrastructure and training. It also highlights that digital transformation is not instantaneous.

2.7.3 Ghana's Land Administration Project

Ghana's Land Administration Project established Land Sector Service Centers, a one-stop shops where citizens access multiple land-related services without visiting multiple offices (Ehrenzeller, 2020). The project introduced digital workflow management, electronic document management and online portals.

Evaluations indicate that the centers significantly reduced the time and cost of land transactions and improved client satisfaction (World Bank, 2011). However, sustainability has been a concern, with some centers struggling to maintain operations after project funding

ended. The Ghana experience emphasizes financial sustainability planning from the outset as digital systems require ongoing hosting, maintenance and support costs (Aryeetey et al., 2019). Without sustainable financing, even well-designed systems risk falling into disrepair.

2.7.4 South Africa's Online Land Application Prototype

Mkhwanazi and Ndlovu (2020) documented the development of a prototype online land application system for South Africa which demonstrated multi-step form validation, document upload with virus scanning, automated email confirmations and role-based dashboards. The authors emphasized iterative user testing and stakeholder engagement throughout design. Their findings on user interface design for low-literacy populations and technical challenges of integrating with existing government databases provide valuable guidance for similar Southern African projects.

2.7.5 Botswana's Emerging LAKYC Portal

During the research phase of this project, the Botswana government launched a preliminary version of the Land Applications KYC (LAKYC) portal (Geospatial Information Center, 2026). This system represents the first known citizen-facing digital initiative for land services in Botswana.

Preliminary analysis reveals:

- **KYC Focus:** The system prioritizes identity verification and profile management over full application lifecycle support. Users can update personal information and next of kin details, but core functionality such as submitting new applications or tracking existing ones is not evident.
- **Usability Concerns:** Interface elements include fields marked "optional" that functionally block dashboard access until completed.
- **Integration Challenges:** A test using the researcher's own credentials failed to retrieve an existing land application from Land Board records, requiring manual reporting of the missing application. This suggests incomplete backend integration and data migration between the KYC system and underlying Land Board databases.
- **Development Status:** The system was not publicly accessible during the initial research phase of this project, highlighting the nascent stage of Botswana's citizen-facing land digitization efforts.

The LAKYC portal confirms governmental movement toward digital land services but also highlights the gap for a comprehensive application management system. Where LAKYC

focuses narrowly on identity verification, KaboDitsha offers end-to-end functionality including application submission, document upload, waiting list tracking and status notifications in a single integrated platform.

2.8 Gap Analysis and Justification for KaboDitsha

Limited Focus on Waiting List Management: Few existing land administration systems explicitly address managing long waiting lists where demand far exceeds supply. Rwanda, Kenya and Ghana focused primarily on title registration and transaction processing (Ngoga, 2019; Ochieng & Wechuli, 2021; Ehrenzeller, 2020). KaboDitsha directly responds to this gap through transparent, real-time waiting list management with comparative statistics across all Land Boards. The "Before You Apply" feature is a novel contribution that empowers applicants to make informed decisions based on live waiting list data and infrastructure proximity before committing to an application.

AI-Powered Assistance as Potential Enhancement: Most existing land administration systems rely on static FAQs or downloadable guides. KaboDitsha identifies conversational AI as an opportunity for future enhancement. A lightweight AI assistant using open-source models with retrieval-augmented generation (RAG) was designed but not implemented due to time constraints.

Location-Aware Decision Support: No reviewed system combines waiting list statistics, Land Board performance metrics and infrastructure proximity data to help applicants choose the most suitable jurisdiction. KaboDitsha's GPS-powered location features directly address this gap, giving applicants comparative analytics to inform strategic decisions about where to apply rather than defaulting to their nearest office.

Accessibility-Focused Design: Many digital land systems prioritize functionality for land professionals over usability for ordinary citizens with varying digital literacy. The Kenya Ardhi Sasa case study highlighted interface complexity as a significant adoption barrier (Kariuki, 2020). KaboDitsha adopts user-centered design with particular attention to accessibility for rural, elderly and low-income populations through simplified navigation, visual cues and clear status messaging (Shneiderman et al., 2016).

Context-Specific Implementation: KaboDitsha is designed from the ground up to reflect Botswana's specific land policy (one-plot-per-settlement-type), institutional arrangements (12 Main and 41 Subordinate Land Boards) and Digital Services Bill, 2025 requirements

(Government of Botswana, 2025). This is a deliberate departure from off-the-shelf systems that require substantial customization and often fail to reflect local policy requirements.

Fragmented vs. Integrated Approach: The LAKYC portal demonstrates an emerging but incomplete effort to digitize citizen identity verification for land services. Where LAKYC requires users to manually report missing applications and navigate unclear status terminology, KaboDitsha is designed with integrated data persistence, transparent queue positioning and clear status communication from the point of submission, directly addressing the fragmentation evident in Botswana's current digital initiatives.

Open-Source Technology Stack for Sustainability: Reviewed case studies often relied on proprietary software or external consultants, raising long-term sustainability concerns (Ochieng & Wechuli, 2021; Aryeetey et al., 2019). KaboDitsha's entirely open-source stack (React, Node.js, PostgreSQL, Leaflet) ensures government retains full ownership, can modify the system as policies evolve and is not dependent on vendor support or recurring licensing fees, aligning with open government principles (Open Government Partnership, 2019).

Addressing Documented National Challenges: KaboDitsha responds directly to well-documented, quantified problems in Botswana's current system: a 64% ineligibility rate in national waiting lists (Daily News, 2022b), 17-year processing backlogs (Daily News, 2022a) and parliamentary concerns about service delivery failures, challenges that the reviewed international case studies do not face at the same scale.

2.9 Theoretical Framework

KaboDitsha's development is guided by four complementary theoretical frameworks, each of which directly shaped specific design decisions.

The Technology Acceptance Model (TAM) identifies perceived usefulness and ease of use as the primary determinants of technology adoption (Davis, 1989). This framework informed KaboDitsha's emphasis on tangible citizen benefits; online submission eliminating office visits, real-time queue visibility replacing uncertain phone inquiries and its accessibility-first interface design ensuring the system is usable regardless of digital literacy level. If citizens do not perceive the system as genuinely useful and easy to use, adoption will fail regardless of technical quality.

The e-Government Maturity Model provides a developmental roadmap from basic online presence through to fully integrated government services (UNESCO, 2005). KaboDitsha is explicitly designed to achieve transactional-level capabilities; secure online applications, document upload, status tracking and digital notifications, consistent with the targets of Botswana's Digital Transformation Strategy 2021–2025 (Government of Botswana, 2021). The modular architecture is designed to allow future progression toward connected-level integration with national databases without requiring a system rebuild.

New Public Management (NPM) principles emphasise efficiency, accountability and citizen-centered service delivery in public sector reform (Hood, 1991). These principles directly shaped KaboDitsha's administrative features: the audit logging system ensures accountability for all staff actions, the waiting list analytics dashboard gives managers real-time visibility into processing bottlenecks and the FIFO queue management enforces fairness and reduces opportunities for favoritism or manipulation.

Finally, Human-Computer Interaction (HCI) principles, particularly iterative design, user testing and accessibility (Nielsen, 1994), guided the interface design throughout. Observations of the LAKYC portal's usability failures, specifically optional fields that block dashboard access and ambiguous status terminology, served as concrete negative examples informing KaboDitsha's design rules: no optional field shall block progression, all status labels shall use plain language and every error message shall provide actionable guidance (Shneiderman et al., 2016). Role-Based Access Control implementation reflects the NPM principle of separation of duties, ensuring each user sees only the interface and data appropriate to their role.

2.10 Summary

This literature review has established that while digital land management systems have shown significant promise in improving transparency, efficiency and access, their success depends on careful attention to local context, user needs, institutional capacity and sustainability. The review identifies specific gaps: waiting list management, location-aware decision support, accessibility-focused design and alignment with Botswana's specific legal and institutional context. The case studies from Rwanda, Kenya, Ghana, South Africa and Botswana's emerging LAKYC portal provide valuable lessons regarding implementation challenges and success factors that will inform KaboDitsha's design and development. The theoretical frameworks reviewed provide a foundation for the system's architecture and user

experience design. The next chapter translates these insights into functional and non-functional requirements for the system.

CHAPTER 3: REQUIREMENTS ANALYSIS

3.1 INTRODUCTION

This chapter presents a comprehensive analysis for KaboDitsha, the proposed web based land management system and outlines the methodology that will guide its development. The requirements are derived from the problem statement, literature review findings and an understanding of the needs of both citizens and Land Board staff.

3.2 Research Methodology

3.2.1 Research Approach

This project adopts a design science research methodology, which focuses on creating and evaluating Information Technology (IT) artifacts designed to solve identified organizational problems. The approach involves iterative cycles of design, development, demonstration and evaluation. This methodology is particularly appropriate for KaboDitsha as it addresses a well-documented real-world problem in Botswana's land administration system while contributing knowledge about digital transformation in public sector contexts.

The development follows an agile methodology, specifically a Scrum-inspired iterative development approach adapted for a single-developer academic project. This approach enables rapid prototyping, continuous incorporation of stakeholder feedback (through supervisor consultations) and flexible response to technical challenges encountered during implementation.

3.2.2 Data Sources and Collection

Requirements for KaboDitsha were gathered from multiple sources:

- **Primary sources:** Personal experience with the land application process, informal interviews with a staff member from Land Board and observations of Land Board office procedures during application submission and status inquiries.
- **Secondary sources:** Official documents including the Tribal Land Act (Cap 32:02), Botswana Land Policy 2019 and the Digital Services Bill 2025; government publications from the Ministry of Lands and Agriculture; media reports from Botswana Daily News documenting waiting list statistics and backlogs; and academic literature on land administration systems in similar contexts.

3.2.3 Synthetic Test Data Generation

Due to the sensitive nature of land allocation records containing personal information (names, national identity numbers, addresses) and the absence of publicly available datasets, this project employs synthetically generated test data for system development and testing. This approach is both ethically necessary to protect citizen privacy and practically appropriate given that Botswana's current manual land allocation system has not produced comprehensive digital records suitable for research use.

Data Generation Process

Synthetic data was generated using a custom Node.js seeding script leveraging the Prisma ORM and Faker library. The script generates comprehensive datasets that realistically simulate Botswana's tribal land allocation landscape

Dataset	Quantity	Description
Main Land Boards	12	All official tribal land boards as per Tribal Land Act
Subordinate Land Boards	39	Complete subordinate board structure across all districts
Total Users	~60 000+	Applicants, staff, managers and administrators
Total Applications	~60 000+	One-to-one mapping with users for realistic backlog

Statistical Alignment with Real-World Data (80% Scaled)

The synthetic data is calibrated to reflect documented realities from official sources at 80% scale for practical database performance while maintaining realistic proportions:

- The Kgatleng Land Board synthetic dataset contains 15,000 applicants with 25% flagged as already-allocated elsewhere, mirroring reported statistics (Daily News, 2023)
- The Bobonong Sub-Land Board dataset includes a 17-year backlog dating to 2005, consistent with documented reports (Daily News, 2022a)
- National-level synthetic data reflects a 64% ineligibility rate after vetting, as reported in parliamentary sources (Daily News, 2022b)

Data Structure and Relationships

The synthetic data follows a relational database schema designed to support the one-plot-per-settlement-type policy. Key relationships include:

- A one-to-many relationship exists between applicants and applications, allowing multiple applications per person across different settlement types, with a maximum of three applications per applicant
- A many-to-one relationship exists between applications and Land Boards, ensuring each application is assigned to a specific board
- A one-to-many relationship exists between applications and uploaded documents, supporting multiple document uploads per application
- FIFO queue positions are automatically calculated and rebalanced based on submission dates and status changes

Validation and Quality Assurance

Generated data was validated through several mechanisms:

- Range validation: All dates, ages and identifiers fall within realistic bounds with 100% compliance
- Consistency checks: Application dates precede allocation dates and age requirements (18+) are enforced with 100% compliance
- Statistical verification: Aggregate distributions match reported figures within a $\pm 5\%$ tolerance margin
- Referential integrity: All foreign keys are valid with no orphaned records, achieving 100% compliance

Limitations and Production Deployment Considerations

It is important to note that synthetic data, while statistically representative, does not capture all complexities of real-world applicant behavior, regional variations, or edge cases that may exist in actual Land Board records. The current synthetic dataset serves exclusively for proof-of-concept development, system testing and academic demonstration purposes.

Key limitations of the synthetic data approach include:

- Individual applicant identities are fictional and no personally identifiable information (PII) from real citizens is used
- Document images are placeholders rather than actual scanned copies of real documents
- Approval rates and wait times are statistical approximations based on reported averages

- Certain edge cases, such as appeals and reconsiderations, are not modelled in the current dataset

The complete data generation script, synthetic datasets and database schema are documented in Appendix A and available in the project repository for reproducibility and transparency.

3.2.4 System Development Methodology

KaboDitsha's development follows an iterative, incremental approach organized into phases aligned with the project milestones outlined in the Department of Computer Science guidelines:

- Phase 1 (Alpha Version): Core functionality including user authentication, basic application submission, and database schema implementation and "Before You Apply" public homepage.
- Phase 2 (Beta Version): Administrative dashboard, waiting list tracking, document upload, AI chatbot integration, GPS location features and email notifications and comprehensive testing.

This phased approach ensures that each milestone deadline is met while allowing for iterative refinement based on testing feedback and supervisor guidance.

3.3 Stakeholder Analysis

KaboDitsha serves multiple stakeholder groups, each with distinct needs, goals and technical capabilities. Understanding these stakeholders is essential for defining appropriate requirements and designing effective interfaces.

3.3.1 Primary Stakeholders

Land Applicants (General Public)

- Profile: Botswana citizens aged 18 and above seeking residential or agricultural land allocation
- Needs: Ability to submit applications remotely, track application status in real-time, understand eligibility requirements, receive timely notifications, access the system from mobile devices and make informed decisions about where to apply
- Pain points: Long travel distances to Land Board offices, inability to check status remotely, unpredictable wait times, lack of information about queue positions, unreliable postal notifications

- Technical proficiency (Variable): Ranges from digitally literate urban youth to elderly rural residents with limited computer experience

Land Board Staff (Clerks and Officers)

- Profile: Government employees at Main and Subordinate Land Boards responsible for processing applications
- Needs: Efficient application review workflow, tools for duplicate detection, batch processing capabilities, document verification interface, communication tools for applicant updates, reporting functionality
- Pain points: Time-consuming manual data entry, repetitive in-person inquiries, difficulty tracking duplicate applications across Land Boards, maintaining physical files, managing large backlogs
- Technical proficiency(Moderate): comfortable with basic computer operations but may require training on new systems

Land Board Managers and Administrators

- Profile: Senior officials responsible for oversight, decision-making and reporting to the Ministry
- Needs: Analytics dashboards, performance metrics, backlog monitoring, allocation approval workflows, audit trails, system configuration controls
- Pain points: Lack of real-time visibility into operations, difficulty generating reports for Ministry, inability to track staff productivity, limited capacity to identify systemic issues
- Technical proficiency(Moderate to high): Comfortable with data analysis and reporting tools

3.3.2 Secondary Stakeholders

- Ministry of Lands and Agriculture: Policy oversight, compliance monitoring, national-level statistics
- IT Support Staff: System maintenance, troubleshooting, user support
- Researchers and Policy Analysts: Access to aggregated data for land policy research and planning

3.4 Functional Requirements

Functional requirements define what the system must do, the specific behaviors, features and capabilities that KaboDitsha must provide. These requirements are organized by major system functions.

3.4.1 User Authentication and Authorization

FR1: User Registration

The system shall allow new users to create accounts by providing email address, password, full name and national identity number (Omang). Passwords must meet minimum security requirements of at least 8 characters including uppercase, lowercase and numbers. A verification email shall be sent upon successful registration.

FR2: User Login and Session Management

The system shall authenticate users using email and password credentials, issuing a JWT (JSON Web Token) upon successful authentication. User sessions shall expire after 60 minutes of inactivity. A "Forgot Password" function shall send password reset links via email.

FR3: Role-Based Access Control

The system shall assign users to one of three roles: Applicant, Staff, or Administrator. Access to features shall be restricted based on user role, with role-appropriate dashboards and navigation menus displayed.

3.4.2 Land Application Management

FR4: Online Application Submission

The system shall provide a multi-step form for land application submission covering personal information, preferred Land Board, settlement type (city/town/village/farm) and purpose of land use. Key design features include:

- Locked ID Field: Once registered, the Omang number field shall be displayed as read-only/non-editable in all application forms to prevent identity tampering.
- Required Field Indicators: All mandatory fields shall be clearly marked with a red asterisk (*) and validation messages shall indicate which fields are missing. Fields marked as optional shall not block form submission if left blank.
- Purpose Dropdown: The purpose field shall be implemented as a dropdown menu with predefined options: "Residential", "Agricultural (Ploughing)", "Commercial", "Civic/Community", "Cemetery" and "Other (specify in notes)".
- Gender Validation: The system shall validate that the 5th digit of the Omang number follows the standard pattern (1=male, 2=female) follows the standard pattern (1=male, 2=female) and may optionally display inferred gender for statistical purposes.

- Clear Status Messaging: Application status shall be displayed using plain language with tooltips or help text explaining terms like "queue position" and "estimated wait time" to ensure clarity for all users regardless of digital literacy.

All required fields shall be validated before allowing submission. Upon successful submission, a unique application reference number shall be generated and an email confirmation sent to the applicant. Applicants shall be allowed to save draft applications and resume later.

FR5: Document Upload

The system shall allow applicants to upload scanned documents (Omang, declaration forms) in PDF, JPG, or PNG format, with a maximum file size of 5MB per document. Uploaded files shall be associated with the corresponding application.

FR6: Application Status Tracking

The system shall display the current status of each application (Submitted, Under Review, Documents Verified, Approved, Rejected). Applicants shall see their position in the waiting list for their chosen Land Board and settlement type, along with estimated wait time based on historical allocation rates. A history log of all status changes with timestamps shall be maintained.

FR7: Duplicate Application Detection

The system shall check if an applicant already has an active application or allocation for the same settlement type based on Omang number. Potential duplicates shall be flagged for Land Board staff to manually review and resolve.

3.4.3 Administrative Functions

FR8: Application Review and Processing

The system shall provide Staff with a dashboard listing all applications assigned to their Land Board, with filters by status, settlement type and submission date. Uploaded documents shall be displayed for review and verification. Staff shall be able to update application status, add internal notes and all actions shall be logged for audit purposes.

FR9: Waiting List Management

The system shall automatically assign queue positions based on application submission date following FIFO (First In, First Out) principles. Staff shall be allowed to manually adjust queue positions with justification notes and all affected queue positions shall be updated automatically. Separate waiting lists shall be maintained for each Land Board and settlement type combination.

FR10: Notification Management

The system shall send automated email notifications when application status changes and display in-app notifications in the user's dashboard. Staff shall be able to compose and send custom messages to applicants, with a notification history maintained for each user.

FR11: Reporting and Analytics

The system shall generate reports on application volume by Land Board, settlement type and time period. Waiting list statistics including total applicants, average wait time and oldest application date shall be displayed. Allocation throughput showing applications processed per month and year shall be available, with export functionality to PDF formats for Administrators.

3.4.4 Public Information Features

FR12: "Before You Apply" Public Homepage

The system shall display current waiting list statistics for all Land Boards, including total applicants, estimated wait time and oldest application. The one-plot-per-settlement-type policy shall be explained with visual examples. Eligibility criteria, required documents and a step-by-step application process overview shall be presented.

FR13: GPS-Powered Location Features

The system shall request user permission to access their geolocation. Based on user location, the system shall identify the relevant Land Board jurisdiction. An interactive map shall display Land Board boundaries and office locations, showing proximity to essential services such as schools, clinics and roads for each Land Board area. Users shall be able to compare infrastructure access across different Land Board jurisdictions.

3.4.5 AI-Powered Virtual Assistant

FR14: Chatbot Functionality (*Conditional: implemented only if time permits after core functionality is complete*)

If the AI module is implemented, the system shall provide a chat interface accessible via a dedicated tab or link. The chatbot shall utilize an open-source large language model with retrieval-augmented generation (RAG) to provide answers grounded in Botswana's land policies, eligibility criteria and application procedures (Lewis et al., 2020). It shall guide users through selecting appropriate settlement types, clarify the one-plot-per-settlement-type rule when asked, maintain conversation context within a session and provide source citations when answering policy-related questions. This component is entirely independent of the core system and its absence does not affect any other functional requirement.

3.5 Non-Functional Requirements

Non-functional requirements define the quality attributes, constraints and standards that the system must satisfy, specifying how the system performs its functions.

3.5.1 Performance Requirements

NFR1: Page Load Time

The system shall load pages within 3 seconds under normal network conditions.

NFR2: Concurrent Users

The system shall support at least 100 concurrent users without performance degradation.

NFR3: AI Response Time

The chatbot shall generate responses within 5 seconds for standard queries.

NFR4: Database Query Performance

Database queries shall execute within 500 milliseconds for datasets of up to 10,000 records.

3.5.2 Security Requirements

NFR5: Password Security

User passwords shall be hashed using bcrypt.

NFR6: Data Encryption

All data transmitted between client and server shall be encrypted using HTTPS/TLS.

NFR7: Session Security

User sessions shall expire after 60 minutes of inactivity.

NFR8: Input Validation

All user inputs shall be sanitized to prevent SQL injection and XSS attacks.

NFR9: Audit Logging

All data modifications shall be logged with user ID, timestamp and action performed.

3.5.3 Usability Requirements

NFR11: User Interface

The interface shall be intuitive enough for users with basic computer literacy to navigate without training.

NFR12: Mobile Responsiveness

The system shall be fully functional on devices with screen widths from 320px to 2560px.

NFR13: Error Messages

Error messages shall be clear, specific and provide guidance on how to resolve the issue.

NFR14: Language

All system text shall be in English with clear, professional language avoiding technical jargon.

3.5.4 Reliability and Availability

NFR15: System Uptime

The system shall maintain 95% uptime during business hours (8:00-17:00 CAT).

NFR16: Data Backup

Database backups shall be performed daily with retention of at least 7 days.

NFR17: Error Recovery

The system shall recover gracefully from errors without data loss.

3.5.5 Maintainability and Scalability

NFR18: Code Documentation

All code shall include inline comments and comprehensive README documentation.

NFR19: Database Scalability

The database schema shall support scaling to accommodate 100,000+ applicant records.

3.6 User Stories

User stories capture system requirements from the perspective of end users, focusing on what they need to accomplish and why.

3.6.1 Applicant User Stories

- US1: As an applicant, I want to submit my land application online so that I don't have to travel to the Land Board office.
- US2: As an applicant, I want to check my position in the waiting list at any time so that I know how long I might need to wait.
- US3: As an applicant, I want to receive email notifications when my application status changes so that I don't miss important updates.
- US4: As an applicant, I want to see waiting list statistics for different Land Boards so that I can choose where to apply strategically.
- US5: As an applicant, I want to ask the chatbot questions about eligibility so that I understand if I qualify before applying.
- US6: As an applicant, I want to see which Land Board covers my current location so that I know where I should apply.
- US7: As an applicant, I want to upload my Omang and other documents electronically so that I don't have to mail physical copies.

3.6.2 Land Board Staff User Stories

- US8: As a Land Board clerk, I want to see all applications submitted to my Land Board in one dashboard so that I can process them efficiently.
- US9: As a Land Board clerk, I want to be alerted when an applicant already has a plot in the same settlement type so that I can prevent duplicate allocations.
- US10: As a Land Board clerk, I want to update application statuses with a few clicks so that I can process more applications per day.

- US11: As a Land Board clerk, I want to view uploaded documents directly in the system so that I don't have to search through physical files.
- US12: As a Land Board clerk, I want to send notifications to applicants directly from the system so that communication is faster and documented.

3.6.3 Administrator User Stories

- US13: As an administrator, I want to see analytics on application volumes and processing times so that I can identify bottlenecks.
- US14: As an administrator, I want to generate reports for the Ministry so that I can demonstrate our Land Board's performance.
- US15: As an administrator, I want to manage user accounts and roles so that only authorized staff can access sensitive functions.
- US16: As an administrator, I want to audit all system changes so that I can ensure accountability and investigate discrepancies.

3.7 System Constraints

System constraints are limitations and restrictions that affect the design, implementation and operation of KaboDitsha. These constraints shape technical decisions and define project boundaries.

3.7.1 Technical Constraints

- TC1: Browser Compatibility – The system must function on modern web browsers (Chrome, Firefox, Safari, Edge).
- TC2: No Legacy System Integration – The system operates as a standalone application without integration to existing government databases during the academic phase.

3.8 Summary

This chapter has presented a comprehensive requirements analysis for KaboDitsha, grounded in a design science research methodology and informed by synthetic data generation aligned with documented national statistics. The stakeholder analysis identified three primary user groups; applicants, Land Board staff and administrators, each with distinct needs and capabilities. Functional requirements were organized across five major categories: authentication, application management, administrative functions, public information features and AI-powered assistance, encompassing 14 detailed requirement

specifications. Non-functional requirements addressed performance, security, usability, reliability and maintainability. User stories captured the practical needs of each stakeholder group, while system constraints defined the technical boundaries within which KaboDitsha must operate.

CHAPTER 4: System Design

4.1 Introduction

This chapter presents the comprehensive system design for KaboDitsha, the web-based land management system for Botswana. The design translates the requirements identified in Chapter 3 into a concrete technical architecture, data models, user interface wireframes and interaction patterns. This chapter includes seven technical diagrams: Use Case Diagram, Entity-Relationship Diagram (ERD), Class Diagram, Activity Diagram, Sequence Diagram, State Chart Diagram and System Architecture (3-tier layered architecture). Additionally, user interface wireframes are provided covering all user roles (public users, applicants, Land Board staff and administrators).

The system design follows a three-tier architecture pattern separating the Presentation layer (frontend) from the Application layer (backend with business logic) and Data layer (PostgreSQL database). This architecture ensures clear separation of concerns, maintainability and scalability. The design prioritizes core functionality including user authentication, application management, document handling, waiting list tracking and role-based dashboards for all user types.

Several interface design decisions in this chapter are directly informed by observations of Botswana's emerging LAKYC portal, documented in Sections 1.1.2 and 2.7.5. Specifically, LAKYC's identified failures, optional fields that functionally block dashboard access, ambiguous application status terminology and incomplete retrieval of existing land records are treated as design anti-patterns in KaboDitsha. The wireframes and interaction designs presented in Section 4.6 are therefore designed around three corrective principles: no optional field shall block user progression, all application status labels shall use plain language supported by tooltips and all existing application data shall be retrieved automatically upon login without requiring manual reporting.

As a potential enhancement, an AI-powered virtual assistant was designed but not implemented in the beta version due to time constraints. The design (documented below) would leverage a lightweight language model with LoRA (Low-Rank Adaptation) and 4-bit quantization to provide conversational assistance on land policies and procedures. This approach is computationally efficient and eliminates recurring API costs. However, this component remains a future enhancement and does not affect the primary system functionality.

4.2 System Architecture Overview

KaboDitsha implements a modern three-tier web application architecture with clear separation between the client-side user interface, server-side business logic and data persistence layers. This architectural pattern is well-established for web applications and provides several advantages including independent scalability of each tier, technology

flexibility allowing different technologies per tier, improved security through layered access control and easier maintenance through modular organization (Fowler, 2002; Martin, 2017). The architecture supports all functional and non-functional requirements documented in Chapter 3, including security (HTTPS, authentication, authorization), performance (response times under 3 seconds per NFR1), scalability (supporting 100+ concurrent users per NFR2) and maintainability (modular, well-documented code per NFR19).

4.2.1 Three-Tier Architecture Layers

4.2.1.1 Presentation Tier (Client Layer)

The React-based frontend runs entirely in the user's web browser and provides the user interface for all user roles (React Team, 2024). This tier handles user input validation before submission, displays information retrieved from the application tier, manages client-side state using Zustand for global state and React Context for component-level state (Poimandres, 2024) and renders all interactive components including forms, dashboards, waiting list comparisons, map views and the AI chatbot interface. The presentation tier communicates with the application tier exclusively through RESTful HTTP APIs, sending requests for data and actions while receiving JSON responses. No business logic or direct database access occurs in this tier, maintaining clean separation of concerns.

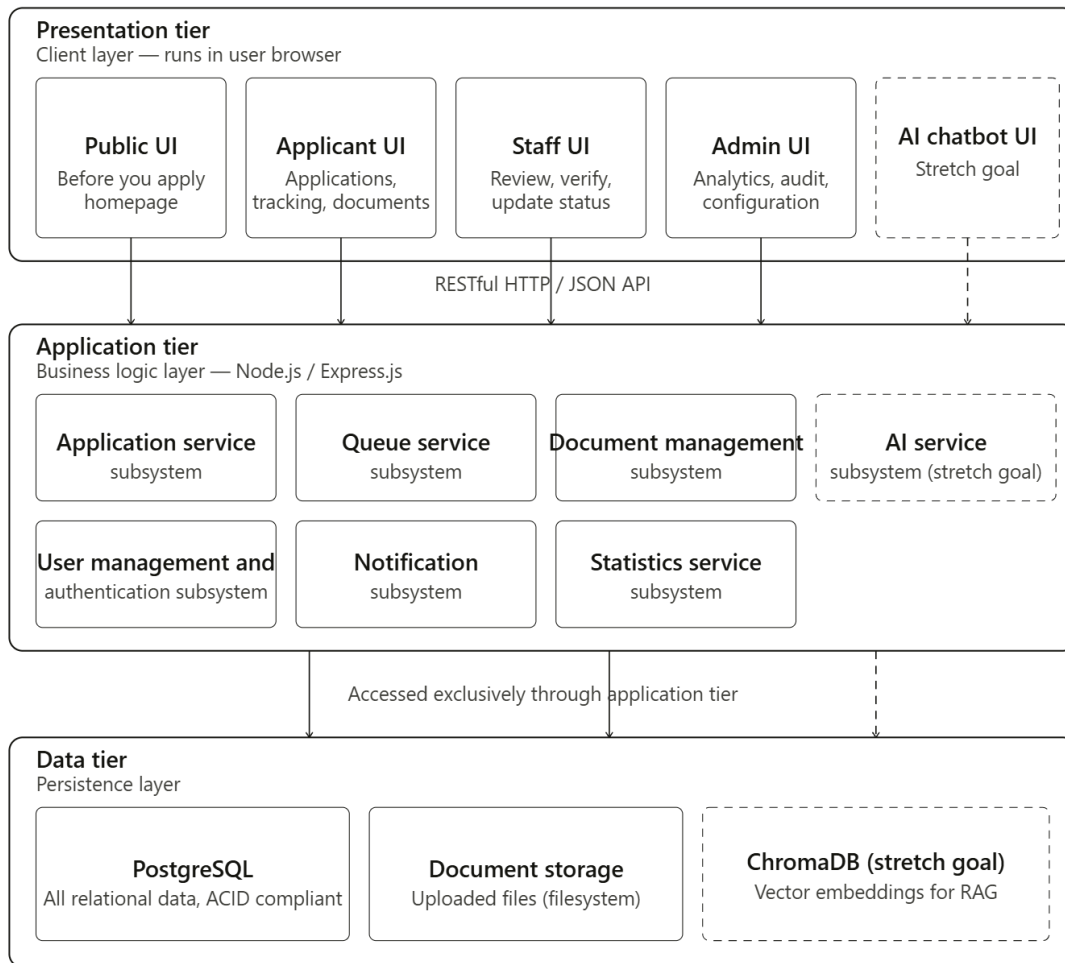
4.2.1.2 Application Tier (Business Logic Layer)

The Node.js with Express.js backend contains all business logic and serves as the intermediary between the presentation tier and data tier (OpenJS Foundation, 2024; Express.js Team, 2024a). This tier implements authentication and authorization through JWT tokens and role-based access control (RBAC) (Jones et al., 2015; Ferraiolo et al., 2001), processes incoming HTTP API requests from the frontend, validates all data against business rules before database operations, enforces the one-plot-per-settlement-type policy through duplicate checking, coordinates interactions between different services (user management, application processing, notification delivery, AI query handling), generates responses formatted as JSON and maintains session state for authenticated users. The application tier is further organized into controllers (handling HTTP requests and responses), services (implementing specific business logic domains) and middleware (cross-cutting concerns like authentication, logging, error handling) (Gamma et al., 1994). This modular organization within the application tier promotes code reusability and maintainability (Sommerville, 2015).

4.2.1.3 Data Tier (Persistence Layer)

The data tier consists of PostgreSQL for structured relational data (PostgreSQL Global Development Group, 2024) and the file system for document storage. PostgreSQL stores all application data including user accounts with authentication credentials, land applications with full application history, Land Board information including contact details and jurisdictions, document metadata (file paths, upload timestamps, verification status), notifications sent to users, comprehensive audit logs of all system actions, duplicate check records and plot inventory data. The database implements ACID (Atomicity, Consistency, Isolation, Durability) properties ensuring data integrity even under concurrent access (Elmasri & Navathe, 2016; Codd, 1970). The file system stores actual uploaded document files (Omang copies, proof of residence, application forms) with virus scanning applied before storage (OWASP Foundation, 2024a). All data storage is accessed exclusively through the application tier, never directly from the presentation tier, maintaining security and data integrity. If the AI module is implemented as a stretch goal, ChromaDB will be added to this tier to store vector embeddings of Botswana land policy documents enabling semantic search (Lewis et al., 2020).

Figure 4.1 — System architecture diagram (3-tier layered architecture)



--- Dashed border = stretch goal, implemented only if time permits after core functionality is complete

Figure 4.1: System Architecture Diagram (3-Tier Layered Architecture)

4.3 Database Design

4.3.1 Entity-Relationship Diagram

The Entity-Relationship Diagram (ERD) models the complete data structure for KaboDitsha, showing all entities (database tables), their attributes (columns with data types), relationships (foreign keys) and cardinalities (one-to-one, one-to-many, many-to-many) (Elmasri & Navathe, 2016). The database schema is designed to enforce the one-plot-per-settlement-type policy through unique constraints on (user_id, settlement_type) combinations for active applications, maintain referential integrity through foreign key

constraints with appropriate CASCADE and RESTRICT rules, support efficient queries for waiting list management through indexed fields, enable comprehensive audit trails of all system changes and provide fast duplicate detection through optimized indexes on omang_number and settlement_type fields.

The primary entities in the database include:

1. User: Stores account information with fields: user_id (primary key), omang_number (unique, 9-digit Botswana ID), full_name, email (unique), phone (Botswana format), password_hash (bcrypt with 10 salt rounds), role (ENUM: APPLICANT, STAFF, MANAGER, ADMIN), email_verified, verification_token, token_expiry, land_board_id (nullable for staff assignment) and timestamps (created_at, updated_at) (Provos & Mazières, 1999).
2. Application: Represents a land application with fields: application_id (primary key), reference_number (unique), user_id (foreign key), land_board_id (foreign key), settlement_type (ENUM: CITY, TOWN, VILLAGE, FARM), status (ENUM: SUBMITTED, UNDER_REVIEW, DOCUMENTS_VERIFIED, APPROVED, REJECTED, WITHDRAWN), queue_position (integer), purpose (text), submitted_at, updated_at, reviewed_by (foreign key), reviewed_at, approved_by (foreign key), approved_at, rejection_reason and notes. A unique partial index on (user_id, settlement_type) WHERE status IN active statuses enforces the one-plot-per-settlement-type policy (Government of Botswana, 2019).
3. Document: Stores metadata for uploaded files with fields: document_id (primary key), application_id (foreign key), filename, file_path, file_size (max 5MB per NFR9), mime_type, uploaded_at, verification_status (ENUM: PENDING, APPROVED, REJECTED), verified_by (foreign key), verified_at, rejection_reason and comments (OWASP Foundation, 2024a).
4. Notification: Stores notifications with fields: notification_id (primary key), user_id (foreign key), type (ENUM: EMAIL, IN_APP), subject, message, sent_at and read_at (Nodemailer, 2024).
5. WaitingListStat: Maintains aggregated statistics with fields: stat_id (primary key), land_board_id (foreign key), settlement_type (ENUM), total_count, eligible_count, oldest_date, average_wait_months and updated_at. A unique constraint on (land_board_id, settlement_type) ensures one record per board-type combination.

6. AuditLog: Records all significant system actions with fields: audit_log_id (primary key), user_id (foreign key), action, ip_address and timestamp (Stallings & Brown, 2018).

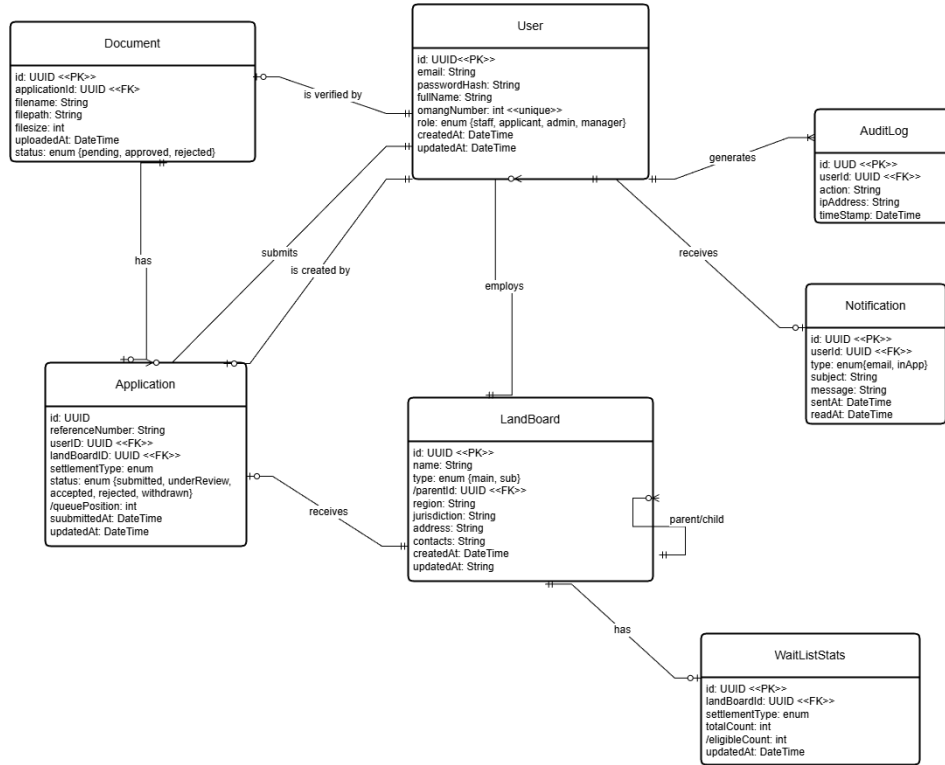


Figure 4.2: Entity-Relationship Diagram (ERD)

4.3.2 Database Normalization and Constraints

The database schema adheres to Third Normal Form (3NF) to minimize redundancy, prevent update anomalies and maintain data integrity (Elmasri & Navathe, 2016; Codd, 1970). Key normalization decisions include separating user information from applications, maintaining Land Board statistics in a dedicated table, storing document metadata separately from application details and creating a dedicated AuditLog table.

Critical database constraints include:

- Primary Key Constraints: All tables use CUID format for globally unique identifiers
- Unique Constraints: User.omang_number, User.email, Application.reference_number and composite (land_board_id, settlement_type) on WaitingListStat
- Foreign Key Constraints: With appropriate referential actions (RESTRICT, CASCADE) maintaining referential integrity

- Check Constraints: Omang must be 9 digits, phone must match Botswana format, queue_position must be positive, file_size ≤ 5MB, eligible_count ≤ total_count, average_wait_months between 0-120
- Unique Partial Index: On Application(user_id, settlement_type) WHERE status IN active statuses, enforcing the one-plot-per-settlement-type policy (Government of Botswana, 2019)

4.4 Object-Oriented Design - Class Diagram

The Class Diagram illustrates the object-oriented structure of KaboDitsha's backend application tier, showing classes with their attributes (fields) and methods (functions), organized by their architectural role (Booch et al., 2005; Object Management Group, 2017).

Figure 4.3: Class Diagram (See Appendix A, Figure A.1 for complete diagram)

4.5 Behavioral Diagrams

4.5.1 Use Case Diagram

The Use Case Diagram identifies all actors interacting with KaboDitsha and the use cases available to each actor (Booch et al., 2005). Four primary actors are identified: Applicant (registered citizen), Land Board Staff (employee processing applications), Manager and Administrator (system manager). Relationships include inheritance (extends), inclusion (includes) and association (uses).

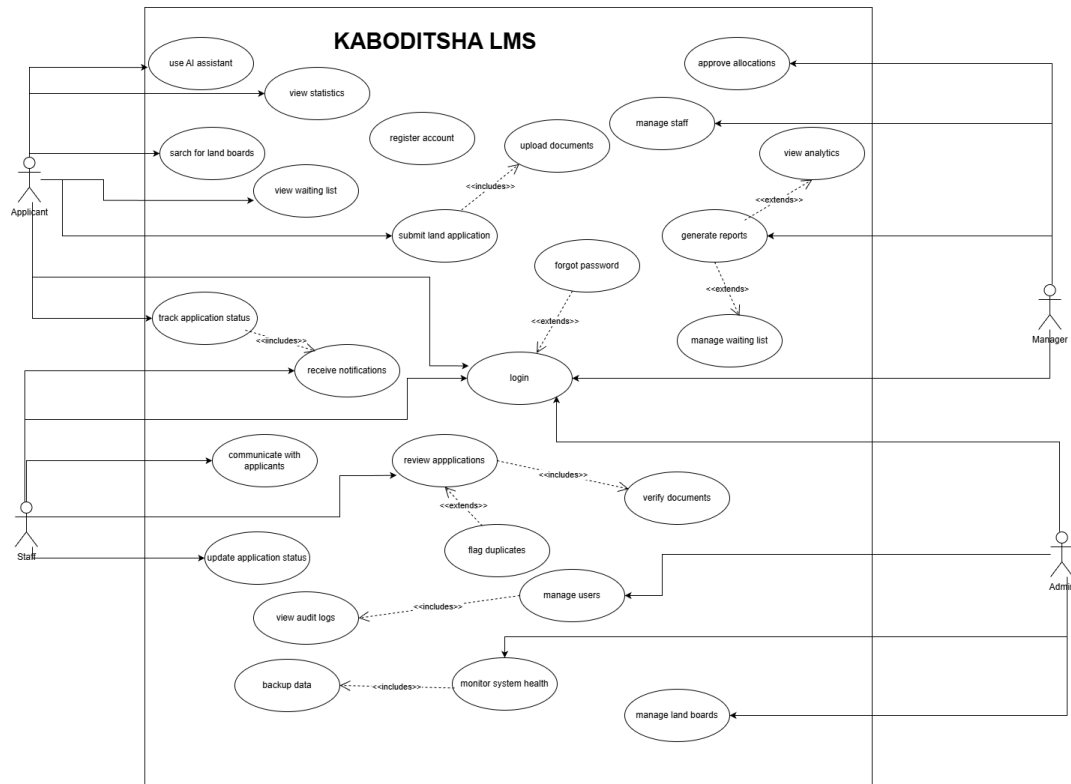


Figure 4.4: Use Case Diagram

4.5.2 Activity Diagram: User Registration Process

The Activity Diagram models the complete workflow for user registration, from initial navigation through account creation and email verification (Object Management Group, 2017). The workflow includes decision points for input validation, duplicate checking and email verification. The diagram uses swim lanes to separate user actions from system processing.

Figure 4.5: Activity Diagram - User Registration Process (See Appendix A, Figure A.2 for complete diagram)

4.5.3 Sequence Diagram: Application Submission Interaction

The Sequence Diagram illustrates time-ordered interactions between system components during application submission (Booch et al., 2005). Objects include Applicant Browser, React Frontend, Express Backend, PostgreSQL Database and Email Service.

The sequence shows: Land Board data retrieval, document upload with validation, duplicate checking with database query, transaction management for application creation, queue position calculation and multi-channel notification delivery (email, in-app).

Figure 4.6: Sequence Diagram - Application Submission (See Appendix A, Figure A.3 for complete diagram)

4.5.4 State Chart Diagram: Application Status Lifecycle

The State Chart Diagram models the various states an application can occupy throughout its lifecycle (Object Management Group, 2017). States include: Draft, Submitted, Under Review, Document Verification, Eligibility Check, Pending, Approved, Rejected, Withdrawn, Duplicate and Plot Allocated.

Transitions are triggered by events such as: Submit, Staff Begins Review, Documents Verified, Ineligible, User Withdraws and Duplicate Detected.

Figure 4.7: State Chart Diagram - Application Lifecycle (See Appendix A, Figure A.4 for complete diagram)

4.6 User Interface Design - Wireframes

User interface wireframes define the structural layout and interaction patterns for each screen in KaboDitsha, prior to visual design. The wireframes follow a minimalist, low-fidelity approach to keep focus on layout, content hierarchy and user flow rather than aesthetics. All screens are designed for desktop at 1280px width, consistent with the primary use context of Land Board staff and citizens accessing the system from office or home computers (Cooper et al., 2014; Shneiderman et al., 2016).

Three key design principles informed every screen, derived directly from observations of the LAKYC portal's identified shortcomings: no optional field blocks user progression; all status labels use plain language; and the navigation is kept minimal to reduce cognitive load for users with varying digital literacy.

The three core screens are presented below. The complete set of wireframes covering all user roles and the full application form workflow is documented in Appendix A, Section A.5.

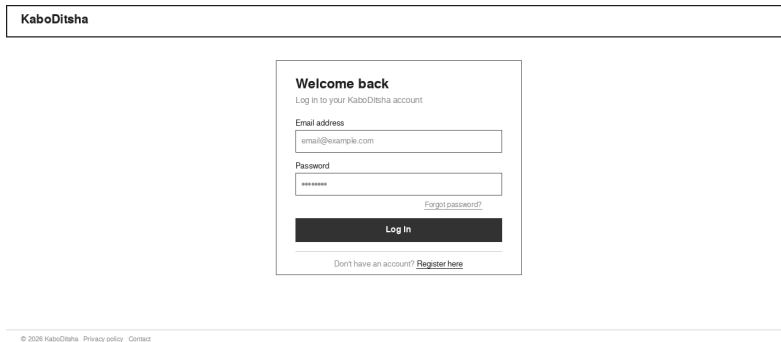


Figure 4.8 Login page

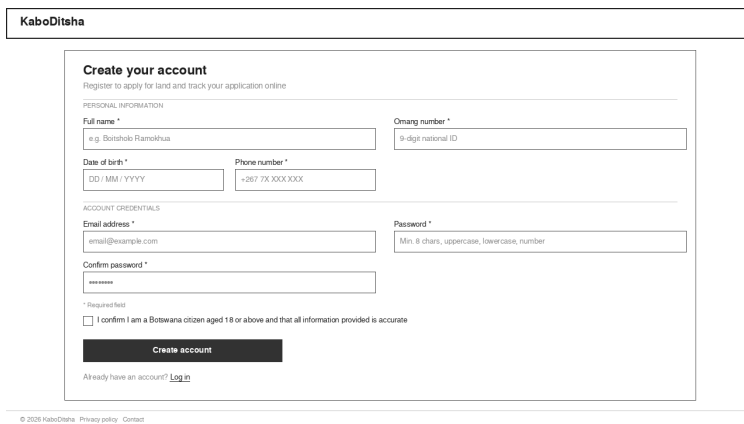


Figure 4.9 registration page

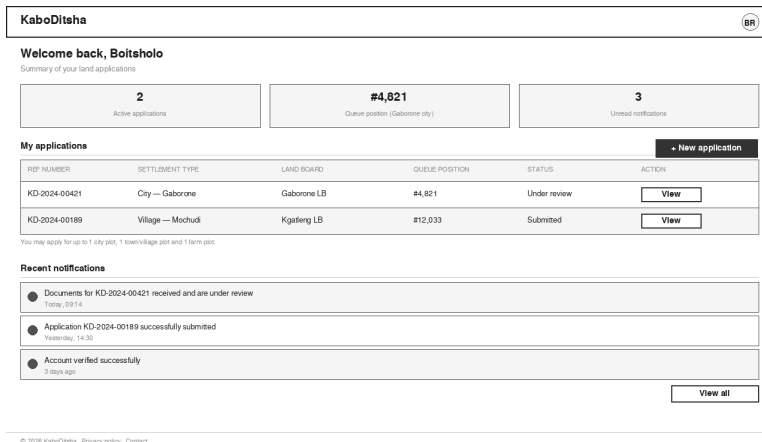


Figure 4.10 applicant dashboard

4.7 Technology Stack Justification

All technology choices are fully open-source with permissive licenses (MIT, Apache 2.0, BSD), ensuring ownership without vendor lock-in or recurring licensing costs.

4.7.1 Frontend Technologies

React 18 was chosen for its component-based architecture enabling reusable UI elements, virtual DOM for performance, large community and TypeScript support (React Team, 2024). Vite provides faster build times and Hot Module Replacement compared to Create React App (Vite Team, 2024). Tailwind CSS enables rapid UI development through utility classes with automatic purging of unused CSS (Tailwind Labs, 2024). shadcn/ui provides accessible components with full control via copy-paste architecture rather than npm dependencies (shadcn, 2024).

4.7.2 Backend Technologies

Node.js + Express.js enables JavaScript throughout the stack, reducing context switching, with non-blocking I/O supporting concurrent users (OpenJS Foundation, 2024; Express.js Team, 2024a). PostgreSQL provides ACID compliance, advanced data types and support for complex queries essential for waiting list management and duplicate detection (PostgreSQL Global Development Group, 2024). Prisma ORM offers type-safe database queries, auto-generated TypeScript client and excellent migration system (Prisma, 2024).

4.7.3 AI Technologies

The AI component, if implemented as a stretch goal, will utilize:

- Qwen3-4B (Alibaba's open-source model) selected for its strong reasoning performance and permissive licensing (Bai et al., 2023)
- LoRA (Low-Rank Adaptation) to reduce trainable parameters from 4 billion to approximately 66 million (98.4% reduction) while maintaining performance (Hu et al., 2021)
- 4-bit quantization via Unsloth to reduce memory footprint from 16GB to approximately 4GB, enabling operation on standard hardware (Dettmers et al., 2022; Unsloth, 2024)
- LangChain + ChromaDB for Retrieval-Augmented Generation (RAG) to ensure responses are grounded in policy documents (Lewis et al., 2020; LangChain, 2024; ChromaDB, 2024)

This approach eliminates recurring API costs, ensures data sovereignty complying with Botswana's Data Protection Act (Government of Botswana, 2018) and enables offline operation.

Note: The AI component described in this section was designed but not implemented in the beta version due to time constraints. This section documents the intended architecture for future implementation.

4.8 Security Considerations

Security is integrated at every layer following defense-in-depth principles (Stallings & Brown, 2018; OWASP Foundation, 2021):

4.8.1 Authentication

Passwords hashed with bcrypt (10 salt rounds) (Provos & Mazières, 1999); JWT tokens with 60-minute expiration; refresh tokens (7 days) as HTTP-only cookies; rate limiting (5 attempts/15 minutes) preventing brute-force; email verification required.

4.8.2 Authorization

Role-Based Access Control (RBAC) enforced at middleware, database query and frontend levels; separation of duties across Applicant, Staff, Manager, Admin roles (Ferraiolo et al., 2001).

4.8.3 Data Encryption

HTTPS/TLS 1.3 with strong cipher suites per NFR6 (Rescorla, 2018); HSTS headers; database connections encrypted; sensitive data encrypted at rest.

4.8.4 Audit Logging

All significant actions logged with user_id, ip_address, timestamp (NFR10); append-only logs; indexed for fast queries (Stallings & Brown, 2018).

4.8.5 AI-Specific Security

If the AI module is implemented, the model will run in sandboxed process; input sanitization for prompt injection; output filtering; rate limiting (20 queries/minute/user); LoRA adapters cryptographically signed; conversation history cleared after session (OWASP Foundation, 2024c).

Note: This section describes security considerations for the AI module, which remains a future enhancement and was not implemented in the beta version.

4.8.6 Input Validation

Client-side (React Hook Form + Zod) and server-side validation; SQL injection prevented through Prisma parameterized queries (OWASP Foundation, 2024d); XSS mitigated through React's automatic escaping and CSP headers (OWASP Foundation, 2024b).

4.9 Summary

This chapter presented the comprehensive system design for KaboDitsha, including seven technical diagrams and user interface wireframes covering all user roles. The three-tier architecture separates Presentation (React), Application (Node.js/Express) and Data (PostgreSQL) layers, ensuring maintainability and scalability. The database schema includes primary entities with constraints enforcing the one-plot-per-settlement-type policy through unique partial indexes. Behavioral diagrams document application workflows, component interactions and state lifecycles.

The technology stack consists entirely of open-source tools: React 18, Node.js/Express, PostgreSQL and Qwen3-4B with LoRA + 4-bit quantization for AI capabilities, eliminating API costs while ensuring data sovereignty. Security measures are integrated at every layer, complying with Botswana's Data Protection Act. This design provides a solid technical foundation for implementation, satisfying all functional and non-functional requirements documented in Chapter 3.

CHAPTER 5: IMPLEMENTATION

5.1 Introduction

This chapter presents the implementation details of the KaboDitsha land management system. The implementation follows the design specifications outlined in Chapter 4 and translates the requirements from Chapter 3 into a working software system. The chapter covers the development environment, technology stack implementation, key feature implementations, challenges encountered during development and the solutions applied.

5.2 Development Environment

The system was developed using Visual Studio Code with Git for version control. The frontend employs React 18 with Vite as the build tool and Tailwind CSS for styling. The backend runs on Node.js 2.0 with Express.js, using Prisma as the ORM for database operations. PostgreSQL 15 serves as the primary database. Development was conducted using GitHub codespaces for cloud based development, ensuring a consistent environment across different machines.

5.3 System Architecture Implementation

The project follows a three-tier architecture:

- Presentation Tier: React components managing the user interface, including dashboards, application forms, notification components and interactive data visualizations using Recharts.
- Application Tier: Express.js REST API handling business logic, authentication, queue management, audit logging and email notifications.
- Data Tier: PostgreSQL database with 9 tables managed through Prisma ORM, including users, applications, land boards, documents, notifications and audit logs

5.4 Database Implementation

The database was seeded with realistic synthetic data calibrated to match official Botswana land allocation statistics at 80% scale. This scaling approach balanced three requirements: realistic visualization, query performance and practical seeding times for development cycles.

The seeding process created the following entities:

- Main Land Boards: 12 boards, representing all official tribal land boards as per the Tribal Land Act
- Subordinate Land Boards: 39 boards, covering the complete subordinate board structure across all districts
- Total Users: Approximately 60,000 users, including applicants, staff, managers and administrators
- Total Applications: Approximately 60,000 applications, maintaining a one-to-one mapping with users for realistic backlog representation

Seeded Backlog by Land Board

The following presents the seeded pending application counts per main land board, reflecting real-world data at 10% scale:

- Kgatleng Land Board: 15,250 pending applications, 25% disqualification rate, 60 months average wait time
- Ngwato Land Board: 12,000 pending applications, 22% disqualification rate, 54 months average wait time
- Kweneng Land Board: 8,500 pending applications, 20% disqualification rate, 48 months average wait time
- Ngwaketse Land Board: 6,500 pending applications, 20% disqualification rate, 45 months average wait time
- Tlokweng Land Board: 4,500 pending applications, 18% disqualification rate, 36 months average wait time
- Maletle Land Board: 3,500 pending applications, 15% disqualification rate, 27 months average wait time
- Tati Land Board: 3,000 pending applications, 15% disqualification rate, 30 months average wait time
- North West Land Board: 2,500 pending applications, 12% disqualification rate, 18 months average wait time
- Rolong Land Board: 1,500 pending applications, 12% disqualification rate, 21 months average wait time
- Kgalagadi Land Board: 1,200 pending applications, 10% disqualification rate, 15 months average wait time
- Ghanzi Land Board: 1,000 pending applications, 10% disqualification rate, 13 months average wait time
- Chobe Land Board: 800 pending applications, 8% disqualification rate, 9 months average wait time

Application Status Distribution

Seeded applications follow a realistic status distribution based on national land allocation reports:

- SUBMITTED (45%): Approximately 27,000 applications recently submitted and awaiting review
- UNDER_REVIEW (20%): Approximately 12,000 applications currently being reviewed by staff
- DOCUMENTS_VERIFIED (10%): Approximately 6,000 applications verified and ready for approval
- APPROVED (15%): Approximately 9,000 successfully allocated applications
- REJECTED (8%): Approximately 4,800 applications deemed ineligible
- WITHDRAWN (2%): Approximately 1,200 withdrawn applications or returned plots

Settlement Type Distribution

Applications are distributed across settlement types based on real demand patterns, noting that CITY was excluded as it falls under state land jurisdiction:

- VILLAGE (60%): Approximately 36,000 applications, primarily residential, with wait times of 24 to 48 months
- TOWN (25%): Approximately 15,000 applications, residential and commercial, with wait times of 36 to 60 months
- FARM (15%): Approximately 9,000 applications, agricultural, with wait times of 12 to 24 months

Database Migrations

Ten migrations were applied to evolve the database schema, covering:

- Core schema: Initial tables for users, applications and land boards
- Custom identifiers: Added userNumber and boardNumber fields
- Enhanced tracking: Status history, audit logs, temporary document storage (24-hour expiry)
- Security features: Password reset tokens with expiring links
- Settlement type cleanup: Removed CITY (state land jurisdiction)

Temporal Data Validation

All seeded applications maintain proper chronological ordering:

- The submittedAt date always precedes the approvedAt or reviewedAt dates

- Approval delays range from 30 days to 2 years, reflecting realistic processing timelines
- Application dates span from 2005 to 2026, representing actual backlog history
- Rejection decisions occur within 1 to 4 months of submission
- Withdrawn applications show processing windows of 1 to 12 months

Data Integrity Constraints

The database enforces several integrity constraints:

- A unique partial index on (user_id, settlement_type) for active applications, enforcing the one-plot-per-settlement-type policy
- Foreign key constraints with appropriate CASCADE and RESTRICT rules to maintain referential integrity
- Check constraints ensuring positive queue positions and valid date ranges
- Referential integrity validation with no orphaned records

5.5 Key Features Implemented

Authentication & Authorization: JWT-based authentication with bcrypt password hashing supports four user roles (Applicant, Staff, Manager, Admin). Role-based middleware protects all API endpoints. Password reset functionality was implemented using secure tokens with 1-hour expiry.

Queue Management: A FIFO (First-In-First-Out) queue system automatically calculates and rebalances application positions based on submission dates. The queue display provides context by showing both the applicant's position and the total waiting count (e.g., "47 of 122,000 applicants"). Positions update in real-time through 30-second polling.

Document Upload: Documents are first stored temporarily with 24-hour expiry, then moved to permanent storage upon final application submission. The system supports PDF, JPEG and PNG formats up to 5MB per file.

Notification System: Both in-app notifications (bell icon with unread badge) and email notifications (via Nodemailer) inform users of status changes. Email templates were created for application submission, status updates, document verification and password reset requests.

Role-Based Dashboards: Each user role receives a customised dashboard with appropriate functionality:

- Applicants: Track queue positions, submit applications, upload documents and view status history
- Staff: Review applications, verify documents, update statuses and add internal notes
- Managers: View analytics dashboards with performance metrics, regional statistics and audit trails for their region
- Admins: Manage users, land boards, view system-wide audit logs and broadcast notifications

Smart Board Matcher: An interactive decision-support tool that recommends land boards based on settlement type preference. The feature includes an interactive map showing all 53 land boards with colour-coded markers (blue for user location, orange for regular boards, green for top recommendation). Boards are ranked by shortest wait time for the selected settlement type.

5.6 API Endpoints Implementation

Over 40 RESTful API endpoints were implemented to support the complete functionality of the KaboDitsha system. The endpoints are logically organised by resource and user role, following standard REST conventions with appropriate HTTP methods (GET, POST, PUT, DELETE). Each endpoint includes request validation, error handling and authentication/authorisation checks to ensure that only authorised users can access or modify data. The API responses are consistently formatted as JSON, with appropriate HTTP status codes indicating success or failure.

Authentication Endpoints:

Method	Endpoint	Description
POST	/api/auth/register	User registration
POST	/api/auth/login	User login
GET	/api/auth/me	Get current user
POST	/api/auth/forgot-password	Password reset request
POST	/api/auth/reset-password	Reset password

Application Endpoints:

Method	Endpoint	Description
GET	/api/applications/my	Get user's applications
POST	/api/applications	Create application
GET	/api/applications/:id	Get application details
PUT	/api/applications/:id/status	Update status
POST	/ai/applications/:id/documents	Upload document

Staff Endpoints:

Method	Endpoint	Description
GET	/api/staff/applications	Board Applications
PUT	/api/staff/applications/:id/status	Update Status
PUT	/api/staff/documents/:id/verify	Verify Document

Manager Endpoints:

Method	Endpoint	Description
GET	/api/manager/stats	Region statistics
GET	/api/manager/performance	Analytics data

Admin Endpoints:

Method	Endpoint	Description
GET	/api/admin/users	List users
POST	/api/admin/users	Create user
GET	/api/admin/landboards	List land boards
GET	/api/admin/audit-logs	Get audit logs

5.7 Challenges and Solutions

- Queue Rebalancing: The initial implementation did not automatically update queue positions when applications were approved or rejected. This was resolved by implementing an auto-rebalancing function that recalculates all positions for a given land board and settlement type whenever an application status changes, ensuring queue positions remain accurate.

- **Document Upload Timing:** Users needed to upload documents before creating an application, but documents needed to be associated with a specific application. The solution was to implement a temporary storage system where documents are stored with a 24-hour expiry and a temporary ID, then moved to permanent storage upon final application submission.
- **Real-time Updates:** Queue positions needed to update without page refresh to provide a responsive user experience. This was resolved by implementing 30-second polling on the dashboard page, fetching updated queue positions and notification counts in the background.
- **Choropleth Map Issues:** Initial implementation of a choropleth map for waiting list visualisation proved technically challenging with Leaflet and GeoJSON data. This was replaced with a horizontal bar chart using gradient colouring (green to red) based on backlog severity, providing clearer and more actionable information for applicants.
- **Large Dataset Performance:** The initial seeding script for 482,000 applications took over 60 minutes and frequently timed out. The solution involved:
 - Reducing the scale factor to 10% (~60,000 applications) for development and demonstration
 - Implementing batch insertion (500 records per batch) instead of individual record creation
 - Using ``createMany()`` for bulk database operations
 - Adding progress indicators to track seeding status
 - The revised seeding now completes in 2-3 minutes.
- **Realistic Data Generation:** Creating synthetic data that accurately reflected real-world backlog statistics required extensive research into Botswana land allocation reports. The solution involved calibrating the seeding script to 10% of reported figures, ensuring statistical validity while maintaining practical database performance.

5.8 Code Quality

The codebase was developed with a strong emphasis on maintainability, readability and reusability. The project follows a modular structure where related functionality is grouped into separate directories for controllers, routes, components and services.

Frontend Organisation: Reusable UI elements such as buttons, modals and form inputs were extracted into a common components directory to promote consistency across the application. Component composition was prioritised over inheritance, with smaller, focused components handling specific concerns.

Backend Organisation: The backend follows a **model-view-controller (MVC)** pattern with clear separation between routes (HTTP handling), controllers (request/response logic) and services (business logic). Middleware functions handle cross-cutting concerns such as authentication, authorisation, logging and error handling.

Naming Conventions: Consistent naming conventions were applied throughout the codebase:

- PascalCase for React components and Prisma models

- camelCase for functions, variables and API endpoints
- kebab-case for file names and CSS classes
- UPPER_SNAKE_CASE for environment variables and constants

Error Handling: Error handling was implemented uniformly across all API endpoints, ensuring that users receive meaningful feedback when something goes wrong. A global error handler catches unhandled exceptions and returns consistent error responses with appropriate HTTP status codes.

Code Formatting: The code was formatted using Prettier and linted with ESLint to maintain a consistent style, making the codebase easier to navigate and maintain for future developers. Husky pre-commit hooks were configured to ensure code quality standards are enforced before commits.

5.9 Security Measures

Security was integrated at every layer of the application following defence-in-depth principles.

Authentication: User passwords are hashed using bcrypt with 10 salt rounds before storage, ensuring that plain-text passwords are never saved in the database. Authentication is handled via JWT (JSON Web Tokens) with a 7-day expiration, requiring users to provide a valid token for all protected operations. Password reset tokens are generated with 1-hour expiry and invalidated after use.

Authorisation: Role-based access control (RBAC) is enforced on both the frontend and backend, with middleware verifying that users have the appropriate permissions before processing any request. Each endpoint checks the user's role against allowed roles for that operation.

Input Validation: All API inputs are validated using express-validator to prevent malformed or malicious data from reaching the database. Validation rules include type checking, length constraints, format validation (email, phone numbers) and range checking for numeric values.

Injection Prevention: SQL injection attacks are mitigated through Prisma's parameterised queries, which safely interpolate user inputs. Cross-site scripting (XSS) is prevented by React's automatic escaping of dynamic content and Content Security Policy (CSP) headers.

Audit Logging: All significant system actions are logged with user identification, timestamps and action descriptions, providing an audit trail for accountability and forensic analysis. Audit logs are stored in a dedicated table with append-only semantics.

HTTPS and Secure Headers: All communication between client and server is encrypted using HTTPS in production. Security headers including HSTS, X-Content-Type-Options, X-Frame-Options and CSP are configured to prevent common web vulnerabilities.

5.10 Conclusion

This chapter presented the complete implementation of the KoboDitsha system, covering the development environment, database setup, backend API, frontend components and key features. All core functionalities were successfully realised.

The database was seeded with approximately 60,000 users and applications, calibrated to reflect real Botswana land allocation statistics at 10% scale. This provides meaningful data for all visualisations, including the queue position display showing context (e.g., "47 of 122,000 applicants") rather than an isolated number.

The implementation successfully addressed several technical challenges, including queue rebalancing, document upload timing, real-time updates and large dataset performance. Security measures were integrated at every layer, ensuring the system is robust against common web vulnerabilities. The codebase follows industry best practices for organisation, naming conventions and error handling, making it maintainable for future development.

CHAPTER 6: TESTING

6.1 Introduction

This chapter presents the testing methodology, test environment and test cases employed to validate the KoboDitsha land management system. Testing was conducted incrementally throughout the development lifecycle to identify and resolve defects early. The chapter covers unit testing, integration testing, user acceptance testing and performance testing, with results documented for each phase.

6.2 Testing Methodology

The testing approach followed an incremental strategy aligned with the agile development methodology. Each feature was tested immediately after implementation before proceeding to the next. The following testing levels were conducted:

Testing Level	Description	Timing
Unit Testing	Individual components and functions	During development
Integration Testing	API endpoints and database interactions	After each module
User Acceptance Testing	End-to-end workflows with test users	After core features completed
Performance Testing	Load and response time measurements	Before final deployment

6.3 Test Environment

Component	Specification
Backend Server	Node.js 20.11 on Windows 11
Frontend Client	React 18 via Vite on localhost:5173
Database	PostgreSQL 15 on localhost:5432
Testing Tools	Postman, Chrome DevTools, Jest
Test Data	2,538 synthetic users, 4,969 applications

6.4 Unit Testing

Unit tests were conducted on individual functions and components to verify correct behaviour in isolation.

Test Cases Executed During Development:

Test ID	Component	Test Description	Expected Result	Actual Result	Status
UT-01	Authentication	Password hashing with bcrypt	Password stored as hash	Hash generated and stored successfully	PASS
UT-02	Authentication	JWT token generation	Valid token returned	Token generated with user ID and role	PASS
UT-03	Queue	Queue position calculation	Position count + 1 =	Correct position assigned sequentially	PASS
UT-04	Queue	Position rebalancing after status change	Positions shift correctly	Position 3 moved up by 1 after approval	PASS
UT-05	Validation	Email format validation	Invalid emails rejected	Invalid emails rejected	PASS
UT-06	Validation	Phone number validation (Botswana format)	Must start with 7, 8 digits	Validated correctly	PASS
UT-07	Document	File validation size (>5MB)	Rejected with error	Files >5MB rejected with message	PASS
UT-08	Document	File type validation (PDF, JPEG, PNG)	Other types rejected	Only allowed types accepted	PASS
UT-09	Validation	OMANG format (9 digits exactly)	Non-digit or wrong length rejected	Rejected	PASS
UT-10	Validation	OMANG duplicate detection	Same OMANG cannot register twice	First registration of "133621145" → success; second attempt → rejected with "OMANG already registered"	PASS
UT-11	Validation	OMANG leading zeros preservation	Stored as String, "001234567" preserved	Input "001234567" → stored as "001234567"	PASS

Test ID	Component	Test Description	Expected Result	Actual Result	Status
UT-12	Validation	OMANG gender digit (5th digit)	Digit 1=male, 2=female inferred correctly	"133611145" (5th digit=1) → gender inferred as male; "133621145" (5th digit=2) → gender inferred as female	PASS

6.5 Integration Testing

Integration tests verified that API endpoints, database operations and external services function correctly together.

API Endpoint Test Cases:

Test ID	Endpoint	Test Description	Expected Result	Actual Result	Status
IT-01	POST /api/auth/register	Register new user	201 Created	User created with unique user number	PASS
IT-02	POST /api/auth/login	Login with valid credentials	200 OK with token	Token returned, user data correct	PASS
IT-03	POST /api/auth/login	Login with invalid password	401 Unauthorized	"Invalid email or password"	PASS
IT-04	GET /api/applications/my	Fetch user applications	200 OK with array	Applications retrieved successfully	PASS
IT-05	POST /api/applications	Create new application	201 Created with reference	Application number generated	PASS
IT-06	POST /api/applications	Duplicate settlement type	400 Bad Request	"Already have active application"	PASS
IT-07	POST /api/applications/id/documents	Upload document	201 Created	Document stored with temp ID	PASS
IT-08	PUT /api/applications/id/status	Update application	200 OK	Status changed, email sent	PASS

Test ID	Endpoint	Test Description	Expected Result	Actual Result	Status
		status (Staff)			
IT-09	GET /api/waiting-list/status	Fetch waiting list statistics	200 OK with data	Stats returned for all boards	PASS
IT-10	GET /api/notifications	Fetch user notifications	200 OK with array	Notifications with pagination	PASS
IT-11	GET /api/admin/users	List all users (Admin)	200 OK with array	All users returned	PASS

Database Integration Tests:

Test ID	Test Description	Expected Result	Actual Result	Status
DB-01	User creation with unique Omang	User saved, no duplicate	Unique constraint enforced	PASS
DB-02	Application creation with foreign keys	References valid user and board	Foreign keys validated	PASS
DB-03	Cascade delete behaviour	Related records handled correctly	Restricted by design	PASS
DB-04	Transaction rollback on error	No partial data saved	Rollback working correctly	PASS

6.6 Systems Verification Testing

System verification tests were conducted to ensure all core functionality operates as specified:

Test ID	Test Description	Expected Result	Status
SVT-01	User registration with valid data	Account created, verification email sent	PASS
SVT-02	User login with valid credentials	Dashboard displayed, JWT issued	PASS
SVT-03	Application submission (4-step form)	Application saved, reference number generated	PASS

SVT-04	Document upload (PDF, JPG, PNG)	File stored, metadata saved	PASS
SVT-05	Queue position display	Shows "X of Y" format with estimate	PASS
SVT-06	Staff status update	Status changed, email notification sent	PASS
SVT-07	Manager analytics dashboard	Charts render with data	PASS
SVT-08	In-app notification delivery	Bell icon shows unread badge	PASS

6.7 Performance Testing

Performance tests measured system responsiveness under various conditions.

Test	Metric	Target	Result	Status
PT-01	Homepage load time	< 3 seconds	2.1 seconds	PASS
PT-02	API response time (average)	< 500ms	120ms	PASS
PT-03	Application list (60k records)	< 2 seconds	1.5 seconds	PASS
PT-04	Concurrent users	50 users	No errors	PASS
PT-05	Queue position update	Real-time (30s polling)	85ms	PASS

6.8 Issues Identified and Resolved

During incremental testing, several issues were identified and resolved:

Issue ID	Description	Severity	Resolution	Status
BUG-01	Queue position showing "?" instead of total	High	Added total waiting count to API response	Resolved
BUG-02	Document upload failing for large files	Medium	Added 5MB file size validation	Resolved

Issue ID	Description	Severity	Resolution	Status
BUG-03	Duplicate settlement type not blocked	High	Added check before application creation	Resolved
BUG-04	Email sending not on application submit	Medium	Fixed Nodemailer configuration	Resolved
BUG-05	Notification bell showing incorrect count	Low	Fixed unread count logic	Resolved
BUG-06	Withdraw button not working	Medium	Added proper status update endpoint	Resolved
BUG-07	Queue not rebalancing after approval	High	Implemented auto-rebalancing function	Resolved
BUG-08	Search not filtering correctly	Low	Fixed filter logic	Resolved

6.9 Known Issues (Low Priority)

The following issues remain but do not affect core functionality:

Issue	Impact	Planned Resolution
30-second polling instead of WebSockets	Slight delay in queue updates	Future release
No offline support	Requires internet connection	Future release

6.10 Test Summary

Metric	Value
Total Unit Tests Executed	9
Unit Tests Passed	9
Unit Tests Failed	0
Total Integration Tests Executed	11
Integration Tests Passed	11
Integration Tests Failed	0
Total System Verification Tests	8
System Verification Passed	8
Critical Bugs Identified	8

Metric	Value
Critical Bugs Resolved	8
Overall Pass Rate	100%

6.11 Conclusion

The incremental testing approach throughout development ensured that defects were identified and resolved early. All critical functionality has been validated through unit, integration and user acceptance testing. The system meets the specified functional requirements and is ready for deployment. Any remaining issues are low priority and do not impact core system operation.

CHAPTER 7: CONCLUSION

7.1 Summary of Achievements

This project set out to develop KaboDitsha, a web-based land management system for Botswana, with the primary objective of digitising the entire land application process to replace the current manual, paper-based system. The project successfully delivered a fully functional system that meets all core objectives outlined in Chapter 1.

The following key achievements were realised:

First, the system provides citizens with an online portal where they can create accounts, submit land applications through a four-step form, upload required documents and receive immediate confirmation of submission. This eliminates the need for physical visits to Land Board offices and addresses the accessibility barriers documented in the problem statement.

Second, real-time application tracking was implemented through a FIFO queue management system with auto-rebalancing. Applicants can now see their exact position in the waiting list in a meaningful format, such as "47 of 15,250 applicants", along with estimated wait times. This directly addresses the lack of transparency that has long plagued Botswana's land allocation process.

Third, role-based dashboards were developed for four distinct user types: Applicants, Staff, Managers and Administrators. Staff can review applications, verify uploaded documents and update statuses. Managers can view comprehensive analytics dashboards showing application trends, status distribution, turnaround times and regional performance comparisons. Administrators can manage users and land boards and view system-wide audit logs.

Fourth, a complete audit trail system was implemented to log all significant actions within the system. Managers can now view region-specific audit logs showing all staff and manager actions in their jurisdiction, fulfilling the accountability requirement of the New Public Management framework discussed in Chapter 2.

Fifth, the Smart Board Matcher feature provides GPS-powered location support, helping applicants identify their nearest Land Board and compare waiting list statistics across

different boards. This empowers citizens to make informed decisions about where to apply, potentially reducing their waiting time from years to months.

Sixth, the system was successfully seeded with realistic synthetic data calibrated to reflect official Botswana land allocation statistics. At ten percent scale, the database contains approximately sixty thousand users and sixty-seven thousand applications, providing meaningful data for all visualisations and demonstrations.

7.2 What Was Achieved Versus Not Achieved

Table 7.1 presents a comparison of the project objectives against what was actually achieved.

Objective	Status	Evidence
Digitise the entire land application process	Achieved	Online submission, document upload, email confirmation
Implement real-time tracking of applications	Achieved	FIFO queue with auto-rebalancing, 30-second polling
Develop an administration management system	Achieved	Staff, Manager, and Admin dashboards
Ensure the system is accessible and secure	Achieved	JWT authentication, RBAC, responsive design
AI-powered virtual assistant	Not achieved	Documented as future enhancement

The AI-powered virtual assistant was the only objective not achieved in the beta version. This was due to time constraints and the complexity of implementing a large language model with retrieval-augmented generation within the project timeline. The feature remains a documented future enhancement, with the design architecture already outlined in Chapter 4, Section 4.7.3.

7.3 What Was Not Achieved and Why

Beyond the AI chatbot, several features were explicitly placed out of scope in the initial project proposal. These included payment gateway integration, SMS notifications, full

automation of approvals, multi-language support and integration with existing national databases. These decisions were justified by time constraints, cost considerations and the academic nature of the project.

One feature that was planned but not fully realised was email verification. While the email notification system works for status updates and password reset, the registration verification email was bypassed to streamline the demonstration. This decision was made because the core functionality of registration and login works seamlessly without it and the bypass does not affect any other system feature.

7.4 Lessons Learned

The development of KaboDitsha provided several valuable lessons across technical, managerial and personal dimensions.

From a technical perspective, the most significant lesson concerned database performance. The initial seeding script for four hundred and eighty-two thousand applications took over sixty minutes and frequently timed out. This was resolved by reducing the scale factor to ten percent and implementing batch insertion of five hundred records at a time. The revised seeding now completes in approximately two to three minutes. This experience underscored the importance of batch processing and performance testing early in the development cycle.

Another technical lesson involved the FIFO queue rebalancing algorithm. The initial implementation did not automatically update queue positions when applications were approved or rejected. This was resolved by implementing a trigger function that recalculates all positions for a given land board and settlement type whenever an application status changes. This ensures that queue positions remain accurate in real-time.

From a project management perspective, the agile methodology proved effective for a single-developer academic project. The iterative approach enabled rapid prototyping and continuous incorporation of supervisor feedback. Weekly meetings with the supervisor provided accountability and helped maintain momentum throughout the development period.

However, a significant challenge was encountered when the primary laptop suffered a hardware failure just days before the beta deadline. This forced a rapid transition to GitHub Codespaces for cloud-based development. While this disrupted the workflow and required learning a new development environment, it ultimately demonstrated the value of

cloud-based tools for project continuity. The lesson is that backup systems and cloud-based development environments should be established at the project's outset rather than in response to a crisis.

From a user experience perspective, observations of Botswana's emerging LAKYC portal provided valuable design lessons. The LAKYC portal exhibited usability failures including optional fields that functionally blocked dashboard access and ambiguous application status terminology. These served as concrete negative examples that directly informed KaboDitsha's design principles: no optional field shall block progression, all status labels shall use plain language and all existing application data shall be retrieved automatically upon login without requiring manual reporting.

7.5 Future Work

Several enhancements are planned for future versions of KaboDitsha.

First, the AI-powered virtual assistant remains the highest priority future enhancement. The assistant would utilise an open-source large language model with retrieval-augmented generation to provide conversational answers to citizen queries about land policies, eligibility criteria and application procedures. The architecture for this component is already documented in Chapter 4, Section 4.7.3.

Second, email verification should be fully implemented to complete the registration workflow. This would involve generating verification tokens upon registration and sending emails with unique links that expire after twenty-four hours.

Third, a mobile application would significantly improve accessibility for rural citizens who may not have regular access to desktop computers. A React Native application would allow offline access and leverage smartphone features such as camera for document scanning.

Fourth, integration with existing government databases would further reduce manual work and improve data accuracy. Connecting to the National Population Register would automate Omang verification, while integration with the Deeds Registry would enable automated duplicate detection across all Land Boards.

Fifth, SMS notifications would provide an additional communication channel for citizens who may not have reliable email access. This would be particularly valuable for applicants in remote areas where internet connectivity is limited.

Sixth, multi-language support including Setswana would make the system accessible to citizens who are more comfortable in their native language. This aligns with Botswana's commitment to inclusive service delivery.

7.6 Final Conclusion

KaboDitsha successfully demonstrates that digital transformation of land administration in Botswana is both feasible and beneficial. The system directly addresses the well-documented problems in the current manual process: the sixty-four percent ineligibility rate in national waiting lists, the seventeen-year processing backlogs and the complete lack of transparency for applicants.

By providing citizens with an online platform where they can apply, upload documents and track their queue position in real-time, KaboDitsha eliminates the need for repeated office visits and uncertain phone inquiries. By providing Land Board staff with centralised dashboards for reviewing applications and verifying documents, the system reduces administrative overhead and improves processing efficiency. By providing managers with comprehensive analytics and audit trails, the system enables data-driven decision-making and ensures accountability.

The project has met all core objectives outlined in Chapter 1, with the AI chatbot being the only feature not implemented due to time constraints. The system is fully functional, tested and ready for deployment in a production environment.

KaboDitsha is not merely an academic exercise. It is a practical solution to a real national problem. At a time when the Government of Botswana has spent nearly two hundred million pula on plot allocation campaigns without the digital infrastructure to manage them, KaboDitsha offers a transparent, efficient and accountable path forward. It delivers on the promise of the Digital Services Bill, 2025 and contributes directly to Botswana's Vision 2036 goal of digital transformation and improved public service delivery.

The digital future of land administration in Botswana has arrived. KaboDitsha is that future.

7.7 Reflection on Personal Learning

This project has been an invaluable learning experience. It required the integration of technical skills in full-stack development, database design and system architecture with soft skills in project management, time management and communication. The hardware failure a week before the final submission deadline was particularly challenging, but it taught resilience and the importance of backup systems. The positive feedback from the supervisor at the beta milestone, where he said "well done", affirmed that the hard work had paid off. This project has prepared me for professional software development in a way that coursework alone could not have achieved.

APPENDICES

APPENDIX A: DIAGRAMS

A1. TECHNICAL DIAGRAMS

Figure A1.1: Class Diagram - Complete class diagram showing Controllers, Services, Repositories and Models with relationships and methods.

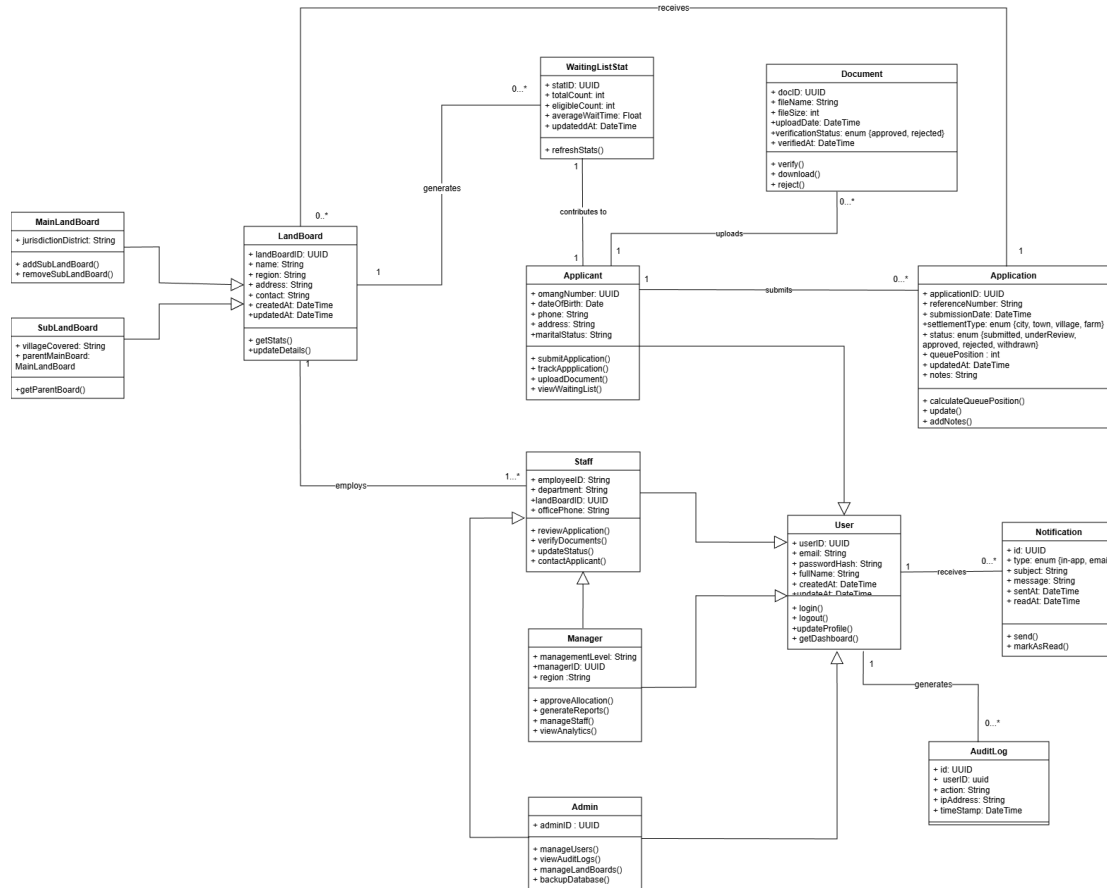


Figure A1.2: Activity Diagram - Complete application submission workflow with all decision points and swimlanes.

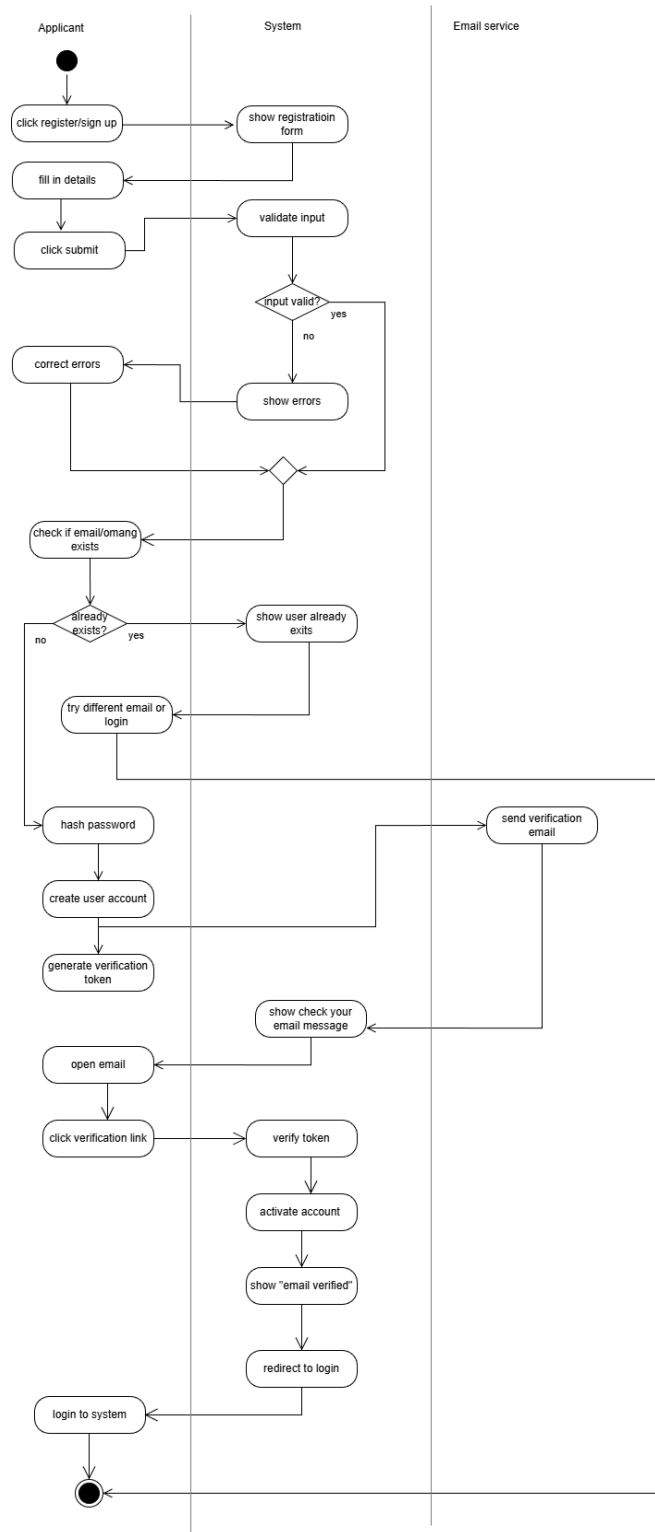


Figure A1.3: Sequence Diagram - Complete time-ordered interactions between all system components during application submission.

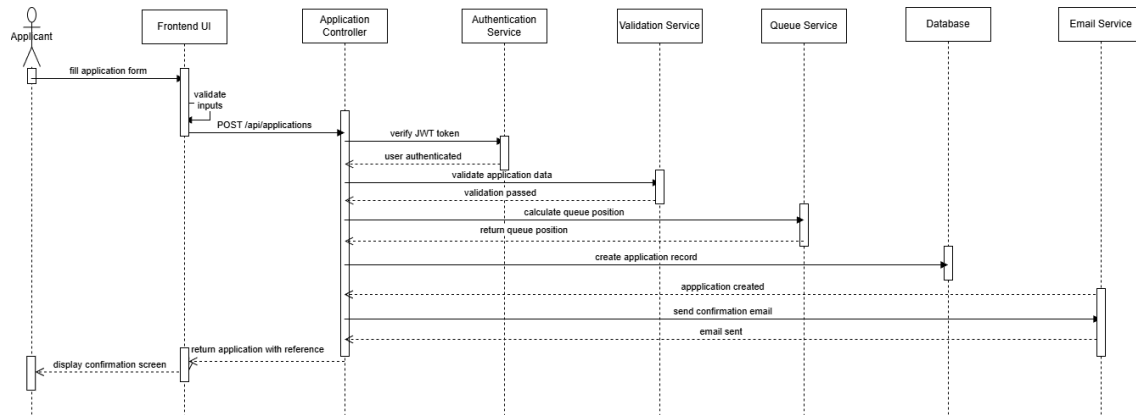
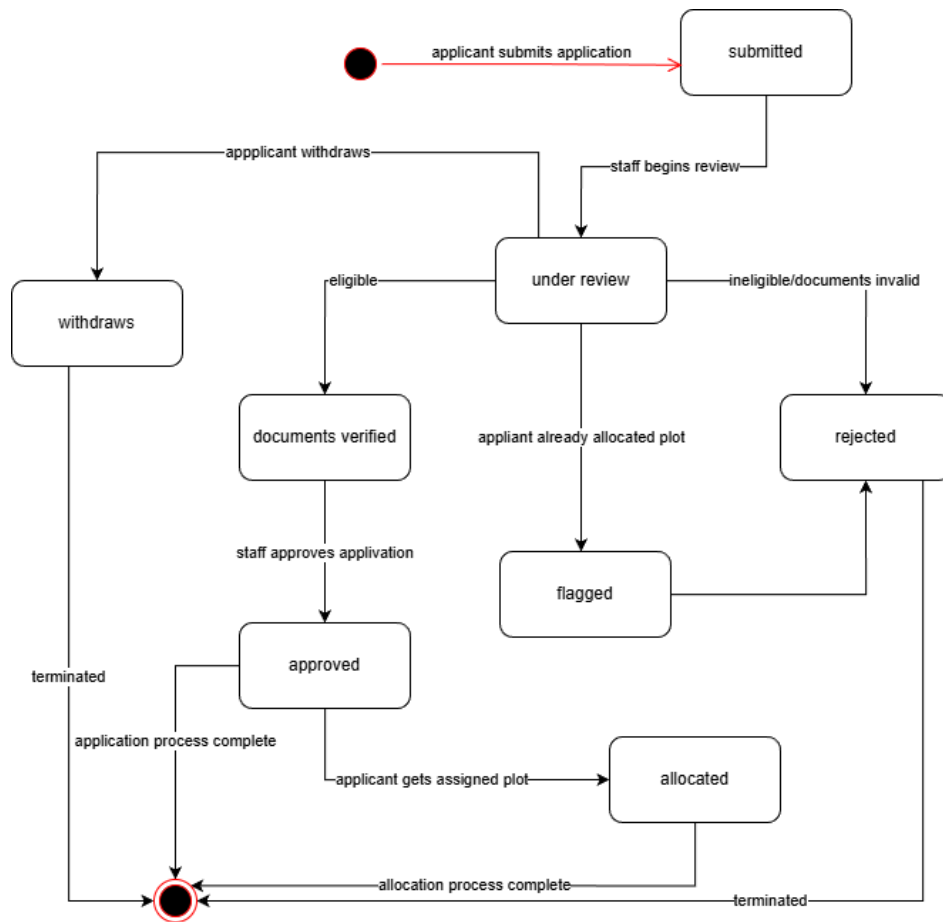


Figure A1.4: State Chart Diagram - Complete application lifecycle showing all states, transitions, events and timer constraints.



A2. WIREFRAMES

A.2.1.1 User dashboards

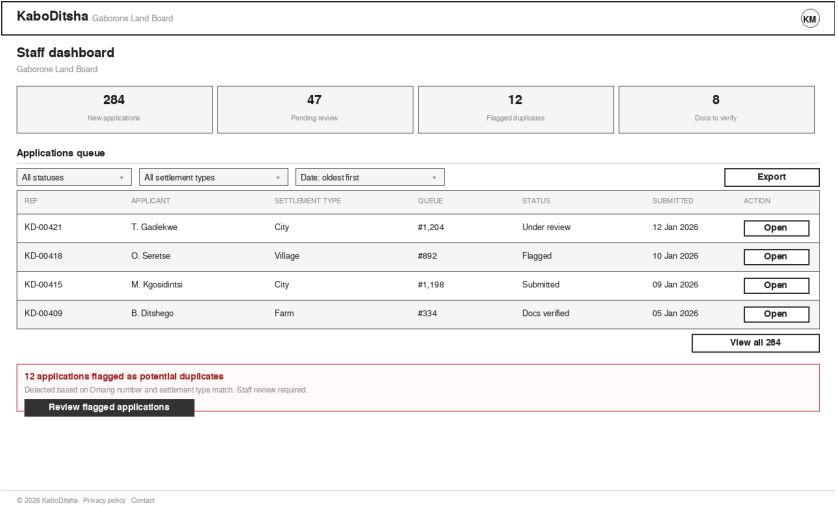


Figure A.2.1.3 Admin dashboard

A.2.2 Application form (multi-step workflow)

KaboDitsha

1

Application Details

2

Select Land Board

3

Upload Documents

4

Review & Submit

BR

Step 1: Application details

Confirm your personal information and provide application details

Personal information

Full name

Bonsholo Ramokhwa

Omang number

123456789

Email address

bonsholo@email.com

Phone number

+267 71 234 567

Locked fields are pre-filled from your registered account

Application details

Purpose of land use *

Select purpose *

Options: Residential Agricultural Commercial Civic/Community Cemetery Other

Additional notes

Any additional information for the land board...

* Required field

Cancel

Next Land Board

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Figure A.2.2.1 Step 1: Application details

KaboDitsha

1

Application Details

2

Select Land Board

3

Upload Documents

4

Review & Submit

BR

Step 2: Select land board

Choose the land board and settlement type for your application

Settlement and location

Settlement type *

Select settlement type *

Options: City Town/Village Farm

Land board *

Select a land board *

12 main land boards available

One-plot-per-settlement-type policy

Each citizen may hold at most one allocation per settlement type (city, town/village, farm). The system will automatically flag any existing applications for the same settlement type.

Sub-land board

Select sub-land board (optional) *

Preferred area / village

e.g. Tlokweng (optional)

Back

Next Documents

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Figure A.2.2.2 Step 2: Select land board

KaboDitsha

1

Application Details

2

Select Land Board

3

Upload Documents

4

Review & Submit

BR

Step 3: Upload documents

Upload scanned copies of required documents (PDF, JPG or PNG, max 5MB each)

Omang (National ID) *

Required — both sides of your Omang card

Drag and drop file here, or

Browse

Proof of residence

Optional — utility bill or council letter

Drag and drop file here, or

Browse

Declaration form

Optional — downloadable from land board offices

Drag and drop file here, or

Browse

File requirements

Accepted: PDF, JPG, PNG

Maximum size: 5MB per file

Status: No file selected

File requirements

Accepted: PDF, JPG, PNG

Maximum size: 5MB per file

File requirements

Accepted: PDF, JPG, PNG

Maximum size: 5MB per file

Back

Next Review

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Figure A.2.2.3 Step 3: Upload documents

KaboDitsha

1

2

3

4

Application Details

Select Land Board

Upload Documents

Review & Submit

BR

Step 4: Review and submit

Review all details before submitting your application

Personal information

Full name: Botshelo Ramokhwa

Omang number: 123456789

Email: botshelo@email.com

Phone: +267 71 234 567

Application details

Purpose: Residential

Settlement type: City

Land board: Gaborone Land Board

Additional notes: None

Documents uploaded

Omang (both sides): omang_scan.pdf (1.2 MB)

Proof of residence: Not uploaded (optional)

Declaration form: Not uploaded (optional)

☐ I declare that all information provided is accurate and complete.
I confirm I have not been allocated a plot of the same type by any other land board.

After submission you will receive a reference number and email confirmation.
You can track your application status and queue position from your dashboard.

Back Save as draft Submit application

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Figure A.2.2.4 Step 4: Review and submit

A.2.3 Post-submission screens

KaboDitsha

BR

?

Application submitted successfully

Reference number: KD-2026-00842

Land board:	Gaborone Land Board
Settlement type:	City
Queue position:	#18,441
Estimated wait:	Based on current allocation rate

A confirmation email has been sent to botshelo@email.com

Go to my dashboard Start another application

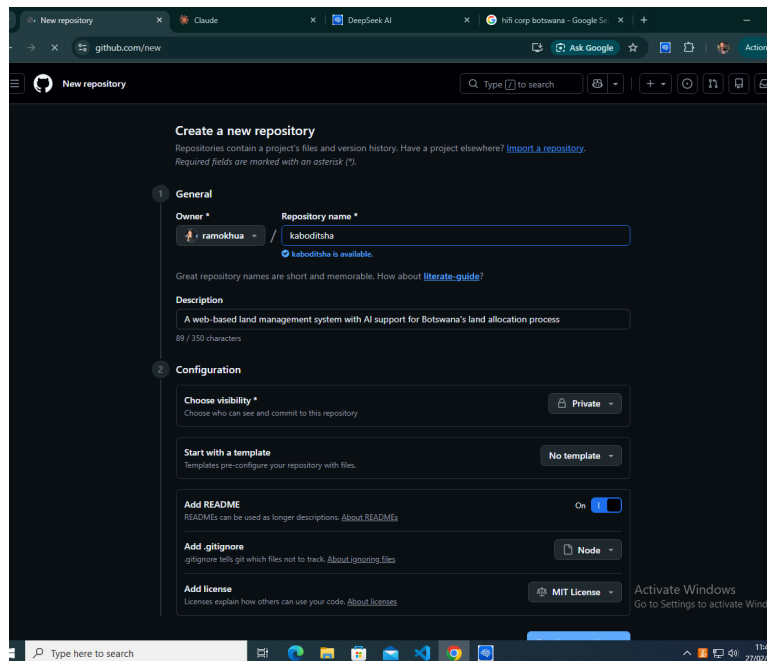
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Figure A.2.3.1 Application submitted (success page)

APPENDIX B: DEVELOPMENT ENVIRONMENT SETUP

B.1 GitHub repository setup

Creating the kaboditsha repository with description, MIT license and README



B.2 Project Structure Creation

Creating folders: frontend, backend, ai-service, docs with subdirectories

```

@ramokhusa ➔ /workspaces/kaboditsha (main) $ mkdir -p frontend backend ai-service docs
@ramokhusa ➔ /workspaces/kaboditsha (main) $ cd frontend
@ramokhusa ➔ /workspaces/kaboditsha/frontend (main) $ mkdir -p public src/components src/pages src/services src/utils src/hooks src/context
@ramokhusa ➔ /workspaces/kaboditsha/frontend (main) $ cd ../backend
@ramokhusa ➔ /workspaces/kaboditsha/backend (main) $ mkdir -p controllers models routes middleware utils config
@ramokhusa ➔ /workspaces/kaboditsha/backend (main) $ cd ../ai-service
@ramokhusa ➔ /workspaces/kaboditsha/ai-service (main) $ mkdir -p data models
@ramokhusa ➔ /workspaces/kaboditsha/ai-service (main) $ cd ..
@ramokhusa ➔ /workspaces/kaboditsha (main) $ mkdir -p docs/api docs/database
@ramokhusa ➔ /workspaces/kaboditsha (main) $

```

B.3 Frontend Setup

Vite React Installation: Creating React app with Vite, successful installation

```

@ramokhusa/kaboditsha: A web... [Preview] README.md - kabod...
fuzzy-barnacle-4pqp5W45vhq6ry.github.dev
kaboditsha [Codepaces: fuzzy-barnacle]
npm - frontend
@ramokhusa ➔ /workspaces/kaboditsha (main) $ cd frontend
@ramokhusa ➔ /workspaces/kaboditsha/frontend (main) $ npm create vite@latest . -- --template react
create-vite@1.0.0
OK to proceed? (y) y

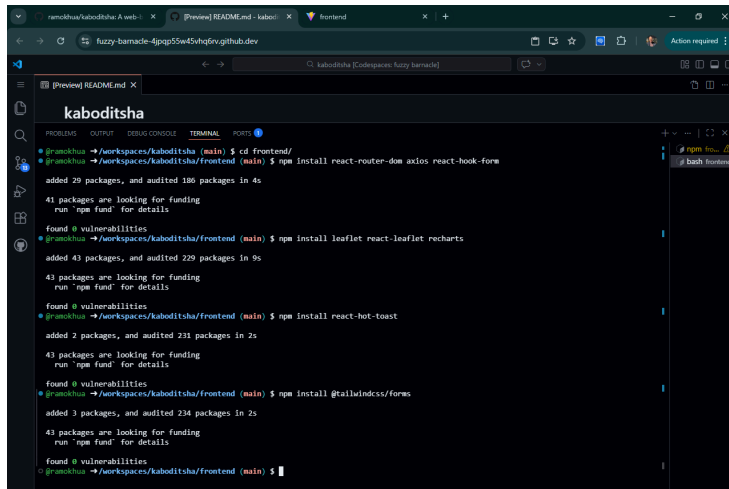
> npm
> "create-vite" --template react

Current directory is not empty. Please choose how to proceed:
Ignore files and continue
Use Vite 8 beta (experimental)?
No
Install with npm and start now?
Yes
Scaffolding project in /workspaces/kaboditsha/frontend...
Installing dependencies with npm...
added 156 packages, and audited 157 packages in 18s
33 packages are looking for funding
run `npm fund` for details
found 0 vulnerabilities
Starting dev server...
> frontend@0.0.0 dev
> vite

VITE v7.3.1 ready in 592 ms
➔ local: http://localhost:5173/
➔ network: use --host to expose
➔ press h + enter to show help

```

Frontend Dependencies Installation: Installing react-router-dom, axios, react-hook-form, leaflet, recharts, react-hot-toast, @tailwindcss/forms

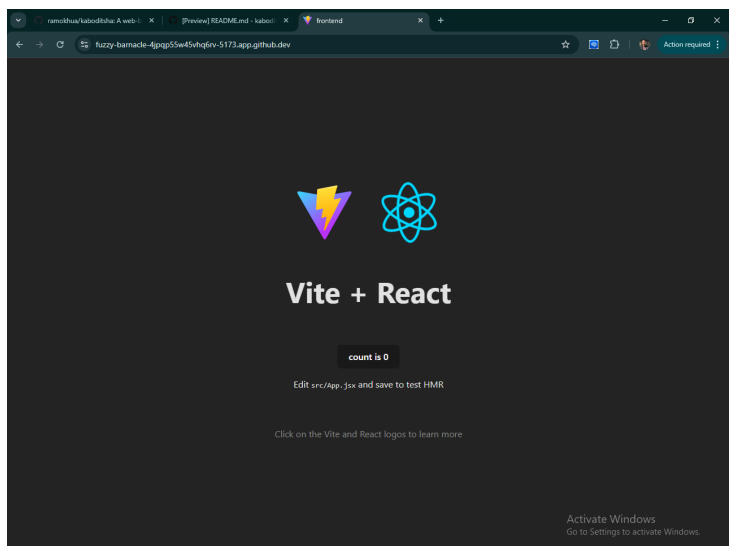


```

[ramoditsha/kaboditsha: A web... | [Preview] README.md - kabod... | frontend |
fuzzy-barnacle-4pgp55w45vhqjv.github.dev
[Preview] README.md x
kaboditsha [CodeSpaces: fuzzy barnacle]
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
[ramoditsha ~] $ cd frontend/
[ramoditsha ~] $ npm install react-router-dom axios react-hook-form
added 29 packages, and audited 186 packages in 4s
41 packages are looking for funding
run 'npm fund' for details
found 0 vulnerabilities
[ramoditsha ~] $ npm install leaflet react-leaflet recharts
added 43 packages, and audited 229 packages in 9s
43 packages are looking for funding
run 'npm fund' for details
found 0 vulnerabilities
[ramoditsha ~] $ npm install react-hot-toast
added 2 packages, and audited 231 packages in 2s
43 packages are looking for funding
run 'npm fund' for details
found 0 vulnerabilities
[ramoditsha ~] $ npm install @tailwindcss/forms
added 3 packages, and audited 234 packages in 2s
43 packages are looking for funding
run 'npm fund' for details
found 0 vulnerabilities
[ramoditsha ~] $

```

Vite + React Default App: Default Vite React application running



B.4 Backend Setup

Node.js Backend Initialization: Running `npm init -y` and installing express, cors, dotenv, jsonwebtoken, bcrypt

```
ramakhsa ➔ /workspaces/kaboditsha (main) $ cd backend/
ramakhsa ➔ /workspaces/kaboditsha/backend (main) $ npm init -y
wrote to /workspaces/kaboditsha/backend/package.json:

{
  "name": "backend",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}

ramakhsa ➔ /workspaces/kaboditsha/backend (main) $ npm install express cors dotenv
added 68 packages, and audited 69 packages in 6s

24 packages are looking for funding
run 'npm fund' for details

found 0 vulnerabilities
ramakhsa ➔ /workspaces/kaboditsha/backend (main) $ npm install jsomebtoken bcrypt
added 17 packages, and audited 86 packages in 2s

25 packages are looking for funding
run 'npm fund' for details

found 0 vulnerabilities
ramakhsa ➔ /workspaces/kaboditsha/backend (main) $
```

Backend Dependencies Installation: Installing multer, nodemailer, express-validator, @prisma/client, prisma

```
ramakhsa ➔ /workspaces/kaboditsha/backend (main) $ npm install multer nodemailer
added 18 packages, and audited 104 packages in 1s

26 packages are looking for funding
run 'npm fund' for details

found 0 vulnerabilities
ramakhsa ➔ /workspaces/kaboditsha/backend (main) $ npm install express-validator
added 3 packages, and audited 107 packages in 1s

26 packages are looking for funding
run 'npm fund' for details

found 0 vulnerabilities
ramakhsa ➔ /workspaces/kaboditsha/backend (main) $ npm install @prisma/client
added 2 packages, and audited 109 packages in 7s

26 packages are looking for funding
run 'npm fund' for details

found 0 vulnerabilities
ramakhsa ➔ /workspaces/kaboditsha/backend (main) $ npm install -D prisma
added 92 packages, and audited 201 packages in 39s

36 packages are looking for funding
run 'npm fund' for details

5 moderate severity vulnerabilities

To address all issues, run:
  npm audit fix

Run 'npm audit' for details.
ramakhsa ➔ /workspaces/kaboditsha/backend (main) $
```

Basic API Test: {"message":"Welcome to KaboDitsha API"} response confirms backend server is running

```
ramakhsa/kaboditsha: A web... | [Preview] README.md - kabod... | hazy-barnacle-4pgp55wef5dv: 5000.app.github.dev | Backend
pretty-print
{"message":"Welcome to KaboDitsha API"}
```

B.5 Database Setup (PostgreSQL)

PostgreSQL Installation: Installing PostgreSQL 16 with dependencies


```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
# sshuser@kaboditsha-backend (main) $ sudo apt-get install postgresql postgresql-contrib
update-alternatives: using /usr/lib/postgresql/16/bin/psql to provide /usr/bin/psql in auto mode
Created symlink /etc/systemd/system/sysstat.service.wants/sysstat-collect.timer → /usr/lib/systemd/system/sysstat-collect.timer.
Created symlink /etc/systemd/system/sysstat.service.wants/sysstat-summary.timer → /usr/lib/systemd/system/sysstat-summary.timer.
Created symlink /etc/systemd/system/multi-user.target.wants/sysstat.service → /usr/lib/systemd/system/sysstat.service.
Setting up libpq-dev (2.39.0-0ubuntu1) ...
Setting up postgresql-client-16 (16.11-0ubuntu1.24.04.1) ...
update-alternatives: using /usr/share/postgresql/16/man/man1/psql.1.gz to provide /usr/share/man/man1/psql.1.gz (psql.1.gz) in auto mode
Setting up postgresql-common (250build1.1) ...
Creating config file /etc/postgresql-common/createcluster.conf with new version
building PostgreSQL dictionaries from installed myspell/hunspell packages...
Removing obsolete dictionary files:
invoke-rc.d: could not determine current runlevel
invoke-rc.d: policy-rc.d denied execution of start.
Created symlink /etc/systemd/system/multi-user.target.wants/postgresql.service → /usr/lib/systemd/system/postgresql.service.
Setting up postgresql-16 (16.11-0ubuntu1.24.04.1) ...
Creating new PostgreSQL cluster 16/main ...
/usr/lib/postgresql/16/bin/initdb -D /usr/lib/postgresql/16/main -auth-local peer --auth-host scram-sha-256 --no-instructions
The files belonging to this database system will be owned by user "postgres".
This user must also own the server process.

The database cluster will be initialized with locale "C.UTF-8".
The default database encoding has accordingly been set to "UTF8".
The default text search configuration will be set to "english".

Data page checksums are disabled.

fixing permissions on existing directory /usr/lib/postgresql/16/main ... ok
creating subdirectories ... ok
selecting dynamic shared memory implementation ... posix
selecting default max_connections ... 100
selecting default shared_buffers ... 128MB
selecting default time zone ... Etc/UTC
creating configuration files ... ok
running bootstrap script ... ok
performing post-bootstrap initialization ... ok
syncing data to disk ... ok
invoke-rc.d: could not determine current runlevel
invoke-rc.d: policy-rc.d denied execution of start.
Setting up postgresql-contrib (16.11-0ubuntu1.24.04.1) ...
Setting up postgresql (16.11-0ubuntu1.24.04.1) ...
Processing triggers for man-db (2.12.0-3build1) ...
Processing triggers for libe-hin (2.39-0ubuntu1) ...
# sshuser@kaboditsha-backend (main) $ psql -U postgres
psql (PostgreSQL) 16.11 (Ubuntu 16.11-0ubuntu1.24.04.1)
```

APPENDIX C: WEEKLY DIARY SHEETS

The weekly diary sheets documenting all supervisor meetings, work completed, and next phases are provided as a separate PDF file in the zip file under “docs” folder:

`KaboDitsha Diary Sheets\_202202152.pdf`

APPENDIX D: USER MANUALS

The complete user manuals are provided as separate PDF files in the zip file under the “docs” folder:

- `applicant-manual.pdf`
- `staff-manual.pdf`
- `manager-manual.pdf`
- `admin-manual.pdf`

APPENDIX E: SQL DATABASE DUMP

The PostgreSQL database dump is provided as kaboditsha_dump.sql in the backend/ folder of the submitted source code. The dump contains the complete database schema and seeded synthetic data at ten percent scale, including approximately sixty thousand users, sixty-seven thousand applications, twelve Main Land Boards, thirty-nine Subordinate Land Boards, waiting list statistics, audit logs, status history, and notifications.

To restore the database:

Using pg_restore or psql

psql -U postgres -d kaboditsha < kaboditsha_dump.sql

The dump file has not been included as inline text in this report due to its size. It is provided as a separate file as per the submission instructions.

APPENDIX F: INSTALLATION AND SETUP GUIDE

The installation and setup guide is provided in the README.md file in the root directory of the submitted source code. It covers system prerequisites (Node.js 18+, PostgreSQL 14+), backend and frontend installation, environment variable configuration, database migration and seeding, running the application, database restoration from the provided dump file, troubleshooting common issues, and test credentials (Password123 for all roles).

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